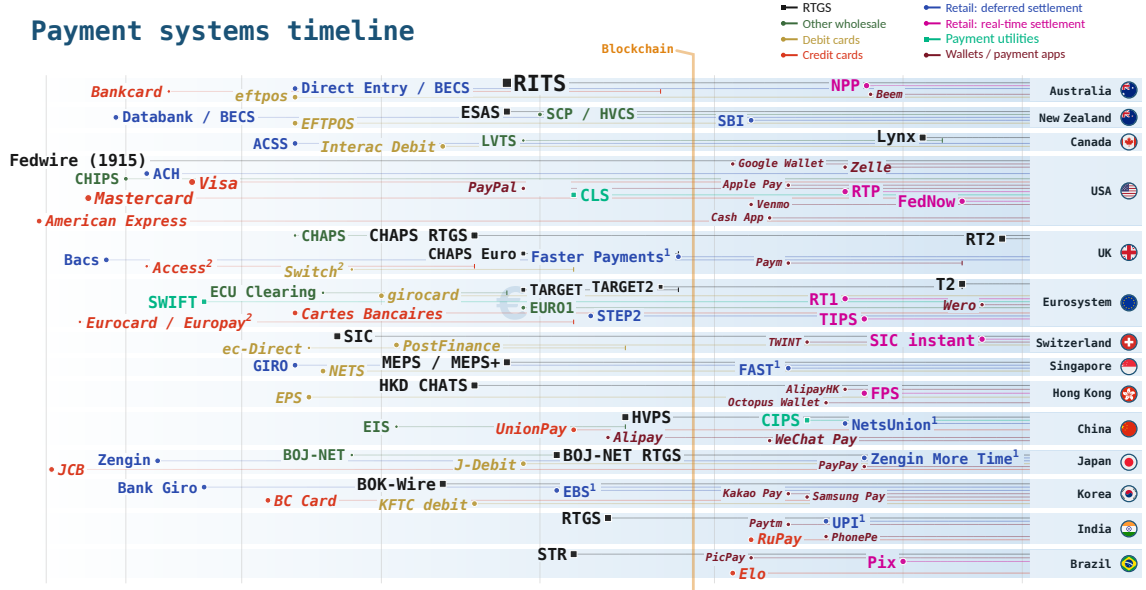
The Role of RITS
in Supporting Settlement in a Tokenised Ecosystem

Reserve Bank of Australia consultation

October 2026

Payment systems timeline



[1]Deferred interbank settlement with instant customer funds. [2]Absorbed into Mastercard: Access (1996), Eurocard/Europay (2002), Switch (2002; UK Maestro until 2011).

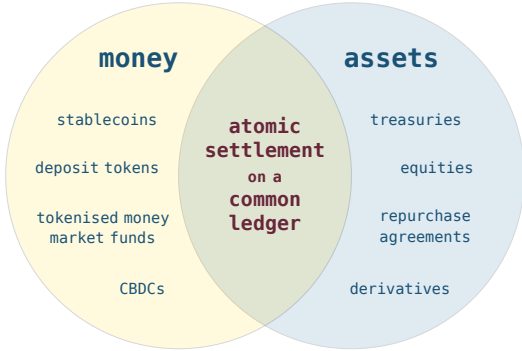
Settlement in a tokenised ecosystem

Tokenised money

Real-time gross settlement (RTGS) systems became widely adopted by central banks from the 1990s. Australia launched RTGS in 1998 with the **Reserve Bank Information and Transfer System (RITS)** as interbank settlement engine. RITS had previously settled government securities, a role that **Austraclear** took over from the RBA in 2002.

Technological limitations restricted RTGS to wholesale payments. Retail RTGS followed in various countries over the next two decades. RITS was enhanced with the 24/7 **Fast Settlement Service (FSS)** in 2018.

Programmable ledgers support new forms of digital money, including deposit tokens and stablecoins. **Project Acacia** explored these systems and their possible links with RITS, helping inform future settlement system design.



Tokenised assets

Central securities depositories (CSDs) immobilised physical certificates and enabled book-entry transfers. Dematerialisation replaced certificates with electronic records. In Australia, **Austraclear** (1984) and **CHESS** (1994) did this for debt securities and equities.

Today, **ASX Group** operates monopoly clearing and settlement through CHESS and Austraclear. Its costly five-year effort to replace CHESS with a **Digital Asset** tokenised platform failed in 2022.

Tokenisation turns money and financial assets into programmable digital objects on a shared ledger. Contract logic governs their issuance, transfer and use. Advantages include **atomicity** of settlement operations, **composability** of automated contracts, and **interoperability** between digital infrastructures.

Tokenisation maps external legal or financial claims into programmable ledger objects. A token can be synchronised to an authoritative off-chain registry – a **digital twin** – or it can represent an authoritative registry entry itself – **native issuance**. Common token interfaces turn different kinds of financial assets into mutually addressable programmable objects.

RITS consultation areas

1. Synchronisation

How RITS/FSS could support DvP settlement of tokenised wholesale assets. Requirements for **synchronising** actions on tokenised platforms with settlement in ESA reserves.

2. At-par exchange

How RITS/FSS could support **at-par exchange** of tokenised private money issued by different institutions, and exchange between traditional bank accounts and tokenised money.

3. Reserve access

How tokenised money issuers might use **central bank reserves** if access were made available, and under what account structures and access models.

4. Tokenised reserves

Which use cases benefit from the issuance of **tokenised central bank reserves**, and the design considerations for their issuance, distribution, funding, and liquidity management.

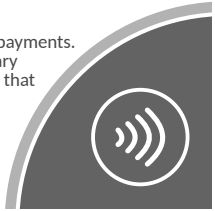
Tokenisation

*Tokenisation in the context of the **RITS consultation** concerns settlement involving **value tokens**.*

Payment Tokens

Substitute payment identifiers

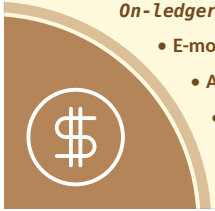
Tokenisation has different meanings within payments. EMV **payment tokens** replace a card's primary account number with a cryptographic token that can be restricted to a device or merchant, limiting misuse if stolen.



***EMV** stands for Europay, Mastercard and Visa (the three biggest card schemes before Mastercard bought Europay). **EMVCo** was established in 1999 to develop common specifications for interoperable chip-card payments.*

Value Tokens

On-ledger value or rights



- **E-money tokens:** e.g. USDC (dollar peg).
- **Asset-referenced:** asset (e.g. gold) or basket.
- **Utility:** issuer's service (e.g. storage credits).
- **Cryptocurrency:** Bitcoin, Ether.

Identity Tokens

ID claims and access rights



OpenID Connect ID Tokens identify who signed in, using JSON Web Token (JWT) format. **OAuth access tokens** authorise actions at a service. **Verifiable credentials** carry signed claims a holder presents repeatedly rather than resupplying documents. **SD-JWT**, a selectively disclosable JWT, reveals only the claims needed.

Context Tokens

Units of data for AI



AI models process text, images, audio and video as tokens. The context window limits tokens used at once. Token counts measure usage; prices reflect compute demand and supply, and may differ for input, output and cached input. Token economics will likely become central to finance as autonomous agents acquire control of value tokens.

Blockchains can record public keys to verify off-chain credentials. Identity tokens can govern who holds or transfers value tokens.

Industry ledgers

Deposit tokens and repo

Service	Instruments and ledger function	Platform	Reported activity
Kinexys Digital Payments *2019*	**Native JPMorgan deposits**: authoritative ledger; treasury payments.	Quorum, Base	**US$1bn+/day** Payments; 2024 report
HQLA^{x} *2019*	**High-quality liquid assets**: tokenised government/corporate-bond entitlements for lending, swaps and repo; ownership changes, securities stay in custody. Canton migration planned.	Corda	**US$5bn intraday repo** First month; Aug 2025
Broadridge DLR *2021*	**Distributed Ledger Repo**: US Treasury repo, intraday repo and collateral pledges; tokenised ownership transfers, securities stay in custody. G7 collateral expansion announced Sep 2026.	Canton	**US$351bn/day** Mean repo; Aug 2026
Partior *2021*	**Interbank payments in tokenised commercial bank money** (USD/EUR/SGD); banks only, 24/7. Founders: DBS, J.P. Morgan, Temasek.	Quorum	**US$1bn+ transferred** Cumulative; Nov 2024
Fnality £FnPS *2023*	**Fnality Payment System**: native sterling wholesale balances on the authoritative ledger, backed by Bank of England funds; earmarking supports conditional payments.	Besu	**Aggregate unavailable** Payment value
Whale (Ant International) *2024*	**Ant group treasury:** blockchain platform connecting partner-bank deposit services; 24/7 intra-group transfers. FX via Kinexys.	Besu	**US$600bn+ processed** Period unstated; Jan 2026
Citi Token Services *2024*	**Citi Token Services for Cash**: tokens record liabilities between Citi branches, not customer deposit claims. 24/7 liquidity transfers, subject to limits; liquidity and working-capital management.	Besu	**~US$1bn/day** Mean flow; May 2026
DBS Token Services *2024*	**DBS (Development Bank of Singapore)**: deposit tokens and programmable 24/7 treasury payments; private Ethereum Virtual Machine-compatible ledger linked to Ant's Whale.	Private EVM	**Aggregate unavailable** Payment value
HSBC TDS *2025*	**Tokenised Deposit Service**: HSBC deposit representations using private distributed ledger technology; 24/7 corporate wallet transfers. Six markets / seven currencies by Aug 2026.	Private DLT	**~US$28bn** Payments; Q1 2026
Standard Chartered *2025*	**Tokenised bank deposits**: SGD/USD balances on Ant's Whale platform; 24/7 intra-group treasury transfers, with additional HKD/CNH capabilities.	Private DLT	**Aggregate unavailable** Payment value
BNY tokenised deposits *2026*	**Bank of New York Mellon deposit mirrors**: private ledger for collateral and margin workflows; authoritative deposit records remain off-chain.	Private DLT	**Aggregate unavailable** Payment value

Industry ledgers

Securities, funds & loans

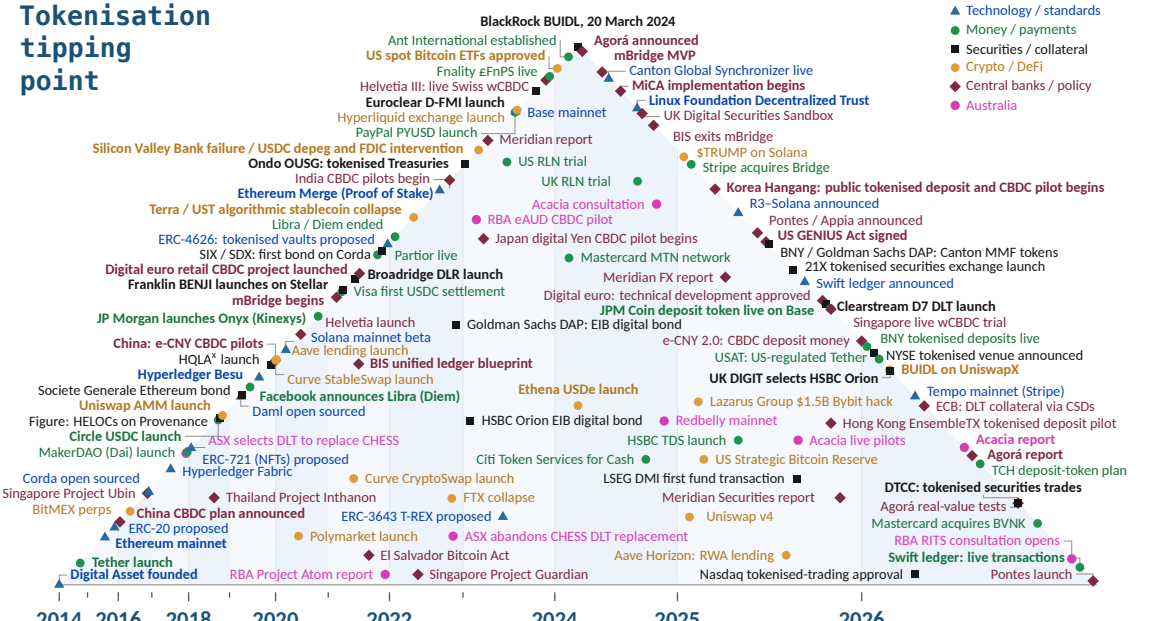
Service	Instruments and ledger function	Platform	Reported activity
Figure *2018*	**Loans, including home equity lines of credit (HELOCs)**; origination, ownership, liens and sales. DART loan register; Figure Connect trading and settlement.	Provenance	**~US$4.3bn** Loan market; Q2 2026
Ant Digital *2018*	**AntChain financial networks**: receivables, trade finance, asset-backed securities, leasing and asset tokenisation; shared records, transfers and financing.	AntChain	**32.4% market share** China BaaS; 2025
Cashlink *2019*	**Crypto-securities register**: bonds and fund units under the German Electronic Securities Act; issuance, ownership records and lifecycle management on public blockchains.	Ethereum, Polygon, Stellar	**EUR1bn+ transactions** Cumulative; Aug 2026
IZNES *2019*	**Fund-unit register**: subscriptions, redemptions, ownership and servicing; money-market, securities, real-estate and private-equity funds. Euroclear is a shareholder.	Hyperledger Fabric	**EUR32bn registered** Fund assets; May 2026
SWIAT *2020*	**Secure Worldwide Interbank Asset Transfer**: bonds, commercial paper and securities lending; appointed registrar maintains the register under the German Electronic Securities Act.	Besu	**EUR600m+ settled** Cumulative; Nov 2025
ADDX *2020*	**Private-market securities**: tokenised funds, equity and debt; issuance, custody and secondary trading on a private permissioned ledger.	Private Ethereum	**SGD2bn+ transactions** Total; Oct 2026
Kinexys Digital Assets *2020*	**Bonds and funds**: Digital Debt Service issuance and servicing; Fund Flow investor registers, capital calls and distributions; intraday repo and tokenised collateral transfers.	Private Ethereum	**US$2–3bn/day** Asset transfers; Nov 2024
Clearstream FundsDLT *2020*	**Digital Transfer Agency**: blockchain fund register for issuance, investor ownership, subscriptions, redemptions and servicing; integrated into Clearstream Fund Services.	Quorum	**EUR11bn+ processed** Digital funds; Jun 2026
SIX Digital Asset Platform *2021*	**Swiss Infrastructure and Exchange**: former SIX Digital Exchange ledger for Swiss cantonal/municipal, supranational and bank bonds and private-share depositary receipts.	Corda	**CHF1.5bn+ issued** Cumulative; 2025 report
Progmat ST *2021*	**Security tokens**: real-estate trust interests, bonds and funds. Full ledger migration from Corda to Avalanche completed Jul 2026.	Avalanche	**JPY248.1bn disclosed** Issued/planned; Jul 2026

Industry ledgers

Securities, funds & loans

Service	Instruments and ledger function	Platform	Reported activity
ibet for Fin (BOOSTRY) *2021*	**Consortium securities ledger** for bonds, shares and fund interests; ownership records and transfers shared across brokers and trust banks.	ibet-Core (EVM)	**JPY133.2bn disclosed** Issued/planned; Jul 2026
Goldman Sachs DAP *2022*	**Digital Asset Platform**: native EIB / Hong Kong Government bonds; Bank of New York Mellon money-market-fund mirrors; bank keeps official books and settlement.	Canton	**Aggregate unavailable** Issuance / fund records
Euroclear D-FMI *2023*	**Digital Financial Market Infrastructure**: bank/supranational Digitally Native Notes, including World Bank; no sovereigns yet. Secondary settlement: Euroclear Bank's existing platform.	Corda	**~EUR1.2bn** Issued; FY2025 report
HSBC Orion *2023*	**Sovereign/supranational bonds**; UK Digital Gilt Instrument (DIGIT) mandate. Luxembourg: Fabric register; public Ethereum mirror tokens have no bond rights.	Hyperledger Fabric	**US$5bn+ issued** Cumulative; Jul 2026
Clearstream D7 DLT *2025*	**Commercial paper / medium-term notes**; European Investment Bank paper as Eurosystem collateral (Jun 2026). Clearstream Banking (Luxembourg): issuer central securities depository.	Besu	**Aggregate unavailable** DLT issuance
SeturionX *2025*	**Tokenised equities, bonds, exchange-traded products and tracker certificates**; formerly BX Digital. Ledger assets / Swiss Interbank Clearing cash, synchronised by the SNB's RTGS link.	Ethereum	**Aggregate unavailable** Exchange turnover
21X *2025*	**Regulated on-chain trading and settlement** of securities and fund tokens; atomic delivery-versus-payment using e-money tokens.	Polygon, Stellar	**~US$1.16m disclosed** 44 trades; Oct 2026
Axiology *2025*	**Lithuania-based debt-securities ledger**: issuance, custody, trading and settlement on a private permissioned network. Cash: euro stablecoins or central-bank money through Pontes.	XRP Ledger (private)	**~EUR17.3m registered** 7 bonds; Oct 2026
LSEG DMI *2025*	**Digital Markets Infrastructure**: private-fund tokenisation, subscriptions and distribution; connects fund managers and investors through LSEG Workspace.	DLT on Azure	**Fundraising live** Since Sep 2025
DTC Tokenization Service *2026*	**Depository Trust Company**: tokenised representations of securities held there; transfers, collateral pledges, lending and delivery-versus-payment.	Besu, Canton	**Limited production** Jul 2026; launch Oct-26

Tokenisation tipping point



AREA 1

Synchronisation

Q1. What DvP synchronisation models (asset lock, central bank reserves lock, or asset and central bank reserves lock) should RITS and/or the FSS be capable of supporting?

Q2. What types of synchronisation arrangements (asset platform-led, third-party or central bank-supported) would best support your use cases?

Q3. To what extent do existing RITS and FSS capabilities and services enable the DvP synchronisation models and synchronisation arrangements that are needed to support current and anticipated use cases?

Q4. What enhancements to the existing services and capabilities are required to support tokenised asset settlement use cases that cannot be adequately supported using existing functionality?

Central-Bank-Money-Settlement-as-a-Service

- **Delivery-versus-payment (DvP).** A trade (asset for money) can record **asset delivery** on an external platform and **payment settlement** in RITS. Each system makes its own transfers final. Removing settlement risk requires ***holding*** one, ***waiting*** for the other to settle, then ***releasing*** the hold. Payment-versus-payment (PvP) describes cross-border trades between currencies settled in different payment systems.

- **RITS already synchronises.** **Austraclear** and **CHESS** (ASX) hold securities and instruct RITS for the cash leg; **PEXA** and **Sympli** reserve funds and settle cash after confirmed lodgement; title registration remains a separate step (near-DvP). **Continuous Linked Settlement** uses its ESA to synchronise RITS with seventeen other central bank systems for PvP FX settlement. The RBA has designated **RITS and CLS** as Australia's two **Systemically Important Payment Systems** (**SIPSs**).

- **Existing RITS links are bespoke integrations.** The Bank of England has explored the alternative of a standardised **operator class** through the **Project Meridian** experiments and is developing an RTGS interface for accredited operators (2026 non-live **Synchronisation Lab**; live service targeted for 2028).

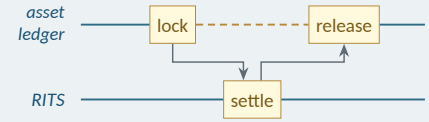
- **Limitations of today's RITS.** RITS is able to reserve funds through reservation batches until the evening cut-off; **the FSS has no reservation facility.** RITS has no common synchronisation API or credit against the asset being purchased.

- **Open question.** Whether the asset leg on a private platform has the statutory finality protection given to RITS *and also to Austraclear* under the *Payment Systems and Netting Act 1998* (PSNA).

DvP/PvP lock and release

1. Seller's asset locked (Austraclear, CHESS)

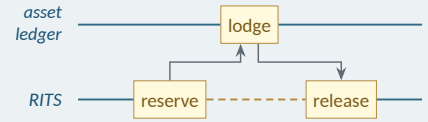
cash final; asset release pending



if the buyer can abort, the lock creates **an effective call option**

2. Buyer's reserves locked (PEXA, Sympli)

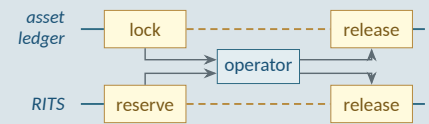
lodgement confirmed; cash release pending



if the seller can abort, the lock creates **an effective put option**

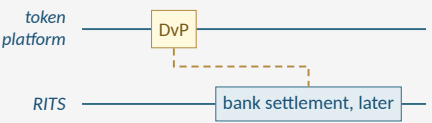
3. Dual lock (Bank of England, Pontes)

operator liveness and timeout risk



4. On-chain DvP (bank-token model)

issuer exposure until the exchange settles



Central bank synchronisation

Synchronisation Operator

Synchronisation Lab

Target: 2028

Operator earmark and release. A **synchronisation operator** instructs **RT2** to earmark the payer's funds and the asset ledger to lock the asset. When both confirm, it instructs both to complete; if either fails the holds are released. Funds move between the participants' own RT2 accounts; the operator holds no central bank money. Intended uses include securities, FX and housing. Prototyped in **Meridian (2023)** and **Meridian FX (2025)**. A Synchronisation Lab opens in 2026 for prospective operators.

Eurosystem Pontes

Launched: Sep 2026

Assets on interoperable ledgers, with launch operators **Clearstream**, **SWIAT**, **Cashlink** and **Axiology**. Connection through **Hash-Link**: the asset settles on their ledger and the cash leg is a direct T2 payment. The second route is a Eurosystem DLT cash-token platform (not yet live). **Pythagore**'s Negotiable EUropean Commercial Paper (NEU CP) market is the first planned user, from end-2026. DLT tokenised reserves mirror T2 cash (the euro RTGS).

RTGS-FMI links

Austraclear / CHESS
Securities–RITS links and batches

Fedwire Securities
Integrated securities–cash settlement

CREST / RT2
CREST cycles; earmarked sterling cash

BOJ-NET
JGB delivery-versus-payment; CSD links

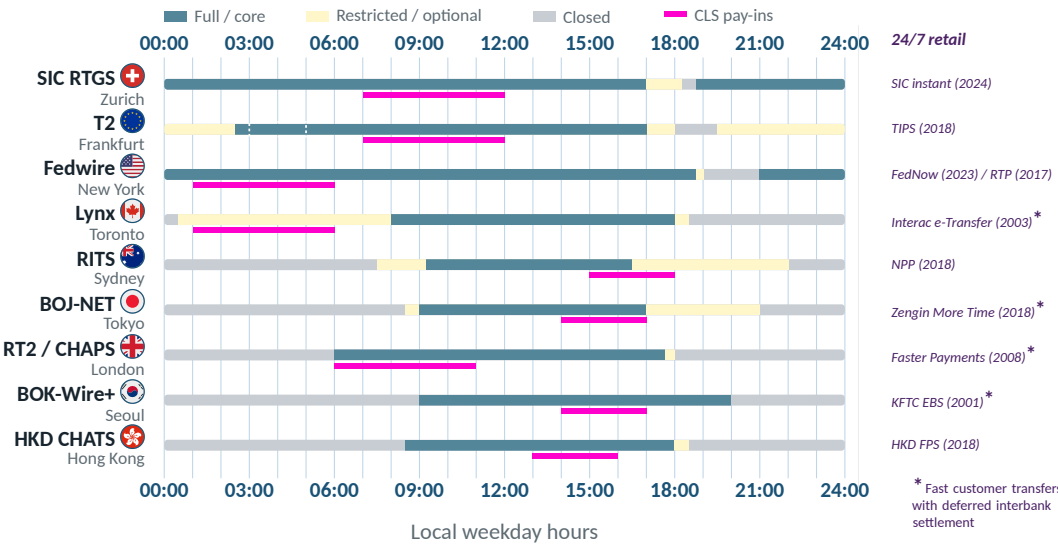
T2S
Common securities–cash platform

SECOM / SIC
Linked securities–cash settlement

CLS multicurrency PvP

Continuous Linked Settlement
75-member user-owned utility created by the international banking industry in 2002. Enables PvP settlement across 18 currencies (over US$8tn daily) within a five-hour window of overlapping RTGS hours (European morning).

RTGS wholesale operating hours



Synchronisation: RITS requirements

Shorter term

- **Reservation lock on the FSS.** Continuous ***hold***, ***observe*** and ***release*** functionality on the settlement account, as in the Eurosystem's **TIPS** (retail payment system). The FSS lacks this.
- **API on RITS/FSS.** ISO 20022 already has reservation messages, `camt.046` to `camt.049`, used in the Eurosystem's **T2** and **T2S**. Operations `lock`, `status`, `settle` and `release`, with callbacks `accepted`, `expiring`, `settled` and `rejected`.
- **Defined operator class.** Authority to instruct a hold on another party's reserves and trigger their release, assuming liability for erroneous release. PEXA and Sympli do this today without infrastructure status. The Bank of England is scoping this class with UK regulators.

Longer term

- **Intraday credit inside the settlement transaction.** T2S-style **auto-collateralisation against the asset being bought** using ISO 20022 messaging. Would require compatible collateral and securities-system interfaces.
- **Admission rules.** RITS could synchronise with different types of operator, in some cases introducing financial stability risks. Formalised platform accreditation criteria must limit operator scope in line with *PFMI* standards.

PayTo does not supply a reservation, only authorisation to a payee to initiate a payment; the payer's bank then sends an ordinary NPP payment, which the FSS settles on receipt. Any hold is on the customer's account at the bank, not on ESA funds.

AREA 2

At-par exchange

Q5. What exchange models (simple exchange, exchange with synchronisation, or DLT exchange) should RITS and/or the FSS be capable of supporting?

Q6. For exchange models requiring synchronisation, what types of synchronisation arrangements (bilateral, third-party, central bank-supported or other) should be supported?

Q7. To what extent do existing RITS and FSS capabilities and services enable the exchange models that are needed to support current and anticipated use cases?

Q8. What enhancements to the existing services and capabilities are required to support exchange use cases that cannot be adequately supported using existing functionality?

Par and tokenised money

- **Monetary singleness.** Australian money is primarily held as bank deposits. Payments between deposit accounts at different banks create interbank obligations that are settled, gross or net, by **transfers between banks' reserve accounts in RITS**. This requirement limits banks' deposit creation and ensures the **fungibility of AUD deposits** at different banks.

- **Par under tokenisation.** A deposit token can circulate within one issuer's liability. Cross-issuer par exchange needs redemption and settlement arrangements, liquidity and scheme rules. Retail holders ordinarily redeem into bank deposits; they do not hold ESAs.

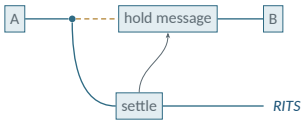
- **Project Acacia recommendation.** Bilateral convertibility becomes harder to scale as issuers and platforms multiply. A **multilateral mint-burn mechanism** linked to interbank settlement would make bank-issued deposit tokens fungible, as RITS ESA settlement does for bank deposits, supporting monetary singleness.

- **Reserves-lock design.** Requires a hold on central bank funds to support **24/7 synchronisation**, or **tokenisation of reserves**. Preferable to designs that can only lock the token or asset on the external platform.

- **Open questions.** *If RITS supports exchange between deposit tokens, should it also support exchange between other regulated money tokens (stablecoins)? Should RITS enforce par value of legally distinct instruments? Which forms of tokenised private money should be eligible for exchange through RITS/FSS? What about a regulated industry ledger, privately operated under national scheme rules?*

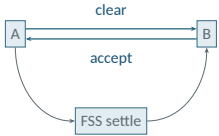
Interbank messaging and payments

Swift Y-copy (HVCS)



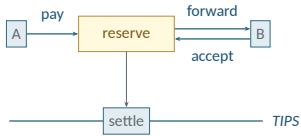
message to the network; ***settle*** *in RITS; then* ***deliver****, no acceptance step. Used for Australian domestic interbank wholesale payments (High Value Clearing System)*

Mesh (NPP)



confirm *between institutions, then* ***settle****; a gateway at every participant used for bilateral clearing of payments before central bank settlement*

Hub with reservation (TIPS)



reserve*,* ***forward****,* ***accept****,* ***settle****; acceptance is due seven seconds from the payer timestamp. Pix/SPI settles before forwarding; corrections use returns*

NPP Australia contracted **Swift** to build and operate a **mesh of bilateral payment gateways (PAGs)** that clear payments between banks before they are settled in the FSS. An alternative to a mesh is a **hub model**, with clearing provided by a central facility like settlement.

Hub model examples include: **TIPS** and **RT1**, **FedNow** and **RTP**, **Pix**, **Faster Payments**, **Zengin**, **SIC instant**, **SPEI**, **FPS**, the **Real-Time Rail**, **UPI**, **PayNow**, **PromptPay**, **DuitNow**, **BI-FAST** and **InstaPay**.

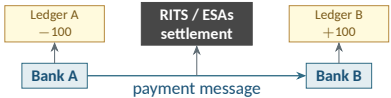
The last six are the members of ***Project Nexus****, which will connect these national retail payment systems through a* ***cross border hub*** *for real-time cross-currency retail payments.*

Australia is the only jurisdiction using a mesh infrastructure design for fast retail payments. This locks in ongoing vendor costs in the order of tens of millions annually. An enhanced FSS would be able to service the NPP under a more efficient hub design, retiring the legacy Swift PAG network.

Messaging and tokens

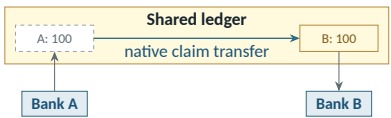
Messages

instruct a trusted intermediary to pay. Banks and the interbank settlement system (RITS) update separate ledgers.



Tokens

Native tokens transfer the recorded claim on an authoritative shared ledger. Mirror and instruction tokens can still require external settlement. Finality depends on the legally authoritative record.

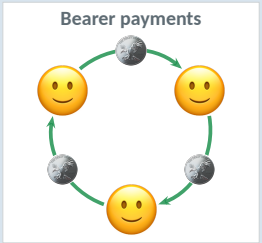


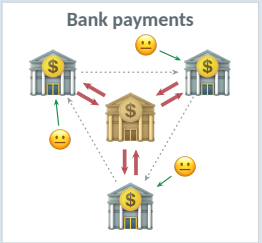
Disintermediation

A token can function as a **cryptographic bearer instrument**, a digital equivalent to physical cash (knowledge of secret keys as possession) which is held and transferred between **self-custodial** users without a bank's involvement. Like physical cash, it may be unrecoverable if lost.

A **retail CBDC** could circulate in this manner without settlement risk, likely reducing public demand for banking services. A **stablecoin** can also circulate in this manner, but is subject to credit and liquidity risk and its par value will depend on user acceptance.

Deposit token interchange preserves the position of banks as token issuers: with tokens being burned and minted in transfer (with interbank settlement support) instead of circulating freely.





ISO 20022 (2004) standardises financial messages. Swift's Cross-Border Payments and Reporting Plus (CBPR+) migration for payment instructions ran Mar-23 to Nov-25, replacing earlier MT messages. Swift is its Registration Authority and publishes the ISO 20022 data dictionary and business-process catalogue.

Money tokens

Central Bank Digital Currency

A direct claim on the central bank (i.e. government), like ESA reserves or physical cash. **wCBDC** (wholesale) settles between banks; **rCBDC** (retail) is held by households and firms. CBDCs may pay interest, as with reserves, or may not, as with banknotes and coins. As central bank money it carries no settlement risk, but as *fiat currency* central bank money it is **unsecured** with no reserve, backed only by taxation.

Stablecoin

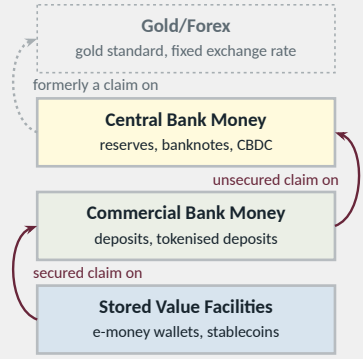
Par value claim **secured** on a pool of backing assets, stablecoins offer bankless self-custody for retail and 24/7 cross-border permissionless payments with near-instant finality. Backing assets may be a mixture of reserves (central bank money), deposits (commercial bank money) and treasury bonds or bills. Regulated stablecoins in most jurisdictions must be 100% backed by HQLA and cannot pay interest. These stablecoins are secured, however the assets securing them are not necessarily convertible to central bank money at par on demand.

An **interest-bearing** stablecoin is often a par-value fund token such as a **tokenised money-market fund**. The split is sometimes drawn as ***savings stablecoins*** (interest-paying) and ***payment stablecoins*** (no interest). A stablecoin fully-backed by central bank money is a reserves-backed digital currency (**RBDC**) or privately-issued "synthetic CBDC".

Deposit token

An **unsecured** (but sometimes government-insured), **interest-bearing** claim to central bank money issued by a commercial bank as a credit extension (*loans create deposits*). May reference a conventional deposit or be issued directly as a token. Deposit tokens are subject to the same prudential regulations as deposits (**Basel III**). Par exchange between issuers (**mint-burn interchange**) resembles payments made from conventional deposits and requires interbank settlement arrangements.

Money hierarchy



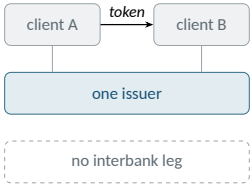
Each layer is a claim on the layer above. Lower layers carry their issuer's credit and liquidity risk; the top layer settles with finality. **RITS ensures that payments made using commercial bank money are anchored to interbank settlement with central bank money.**

Monetary singleness: an AUD$100 payment is accepted as AUD$100 whichever bank issued it. RITS settles interbank payments at par through **Exchange Settlement Accounts (ESAs)**, making Australian-dollar bank deposits fungible across banks.

Token issuance

Single issuer, one token type

token transfer

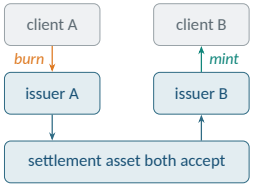


A transfer changes the holder, while the claim remains on the same issuer. "One token" means one token type, which may circulate on several chains.

Original JPM Coin system *(now Kinexys Digital Payments): private network.* ***USDT****: multiple public networks.*

Separate issuers, distinct tokens

Burn-mint interchange

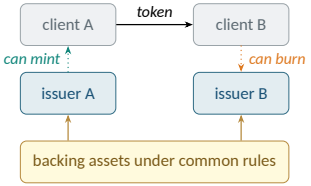


Payment extinguishes A's tokens; B's token issuance coordinated with settlement between issuers. Resembles conventional interbank payments.

Project Hangang*: banks' deposit tokens; interbank settlement in wholesale central bank money (real-value pilot).*

Common token, multiple issuers

AUD+ Scheme Proposal



Multiple legal issuers issue one fungible token under common rules. Backing, redemption and any mutual guarantee require explicit legal arrangements.

M0 / $M*: institutions admitted by protocol governance mint the same $M token against eligible collateral.*

Minting permission alone does not make an entity an issuer: the issuer carries the redemption obligation, regardless of who operates the mint. Distinct tokens are separate issuers' liabilities; exchange is at market prices unless an arrangement supports par exchange. Redemption follows the applicable issuer's terms, including eligibility and fees. A single brand can span several legal entities in one group, each with its own terms. For example, USDC is co-issued by Circle's US and French entities; redemption rights depend on the holder's jurisdiction.

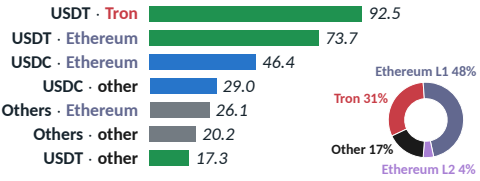
Payment Stablecoin Design Space

	Single-issuer	Multi-issuer
Off-chain reserves	• **USDT**, **Tether** (El Salvador): US$184bn; reserves held and managed by the issuer, with attestations rather than audited accounts. • **USDC**, **Circle**: US$74.9bn; segregated reserves in the Circle Reserve Fund, a registered government money-market fund. EURC EUR0.5bn. • **USDG / Global Dollar**: **Paxos** (Singapore / Europe); 150+ distribution partners share reserve earnings. Ethereum / Solana; US$3.5bn+ circulating, 3 Sep 2026. • **OUSD / Open USD**: **Bridge Building Inc.**; 200+ partners, partner board and shared reserve earnings. Base / Ethereum / Solana / Tempo; live Sep 2026. • **Qivalis**: 37-bank euro stablecoin venture; bank-owned issuer. **DNB** e-money authorisation pending; not issuing.	• **M0 / $M**: US$0.2bn Approved institutions mint against verified off-chain Treasury collateral. Other brands issue tokens backed by $M: **MetaMask mUSD**, **Noble USDN** and the **KAST Dollar**. Protocol rules set eligibility and limits; issuer obligations follow the legal arrangement.
On-chain reserves	• **USDtb**, **Ethena**: US$0.5bn; *reserves are BlackRock BUIDL tokens and stablecoins.* • **USDe**, **Ethena**: US$4.9bn. Holds ETH, BTC and stablecoins *Mixed backing / **delta-neutral hedging**: a US$1 gain on the crypto assets is offset by roughly US$1 lost on short **perpetual futures** and vice versa. Deviations from the peg may be caused by variability in funding costs, liquidations, and imperfect hedges. Direct minting and redemption are limited to approved users.*	• **GHO**, **Aave**: US$0.7bn. Created against excess collateral or in exchange for other stablecoins; Aave governance controls minting rules. • **Sky USDS / Dai**: US$11.5bn. Backed by both crypto assets and conventional financial assets.

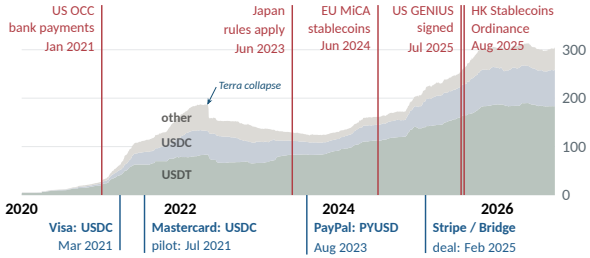
Off-chain reserves: assets in conventional custody. **On-chain reserves**: backing tokens, sometimes with off-chain underlying assets. **Single-issuer**: one issuer group. **Multi-issuer**: independent issuer groups or protocol minters.

Payment stablecoins

Coin · network circulation (US$bn)



USD-pegged supply outstanding (US$bn)



Provider	Initiative	Use cases
Visa	**USDC settlement** (Mar 2021) **Bridge cards** (Apr 2025) **Visa Direct** pilots (Sep/Nov 2025)	**Card settlement:** USDC live seven days a week; EURC pilot; PYUSD / USDG support announced; pilot expanded to nine chains (Apr 2026). **Card spending:** Bridge cards spend stablecoin balances; expanded Mar 2026. **Transfers and payouts:** Visa Direct pilots for stablecoin prefunding and wallet payouts.
Mastercard	**USDC pilot** (Jul 2021) **Settlement expansion** (Jun 2026) **BVNK** acquired (Aug 2026) **SoFiUSD** live (Sep 2026)	**Card settlement:** USDC in select markets; SoFiUSD live, with 24/7 receipts in SoFi accounts; PYUSD, USDG, USDP and RLUSD expansion announced. **Card spending:** spend stablecoin balances. **Transfers and payouts:** Mastercard Move bank withdrawals; Nuvei / Circle merchant USDC payouts announced. BVNK holds, converts and transfers bank money and stablecoins.
Stripe / Bridge	**Checkout** (Oct 2024) **Financial Accounts** (May 2025)	**Checkout:** customers pay stablecoins; merchants receive fiat in their Stripe balance. **Accounts / Bridge:** receive bank or wallet transfers, hold dollar stablecoins, convert between fiat and stablecoins, and pay bank accounts or crypto wallets.
PayPal	**PYUSD** (Aug 2023) / **Xoom** (Apr 2024) **Pay with Crypto** (Jul 2025)	**PYUSD:** Paxos-issued dollar stablecoin for holding, payments and wallet transfers. **Pay with Crypto:** crypto checkout converted via PYUSD; merchants receive dollars. **Xoom:** PYUSD-to-fiat payouts over local payment rails.
Circle	**Circle Payments Network (CPN)**	**Cross-border payments:** Connects financial institutions that convert local currency into stablecoins, transfer them, and convert back for local-currency payouts.

Savings Stablecoin Design Space

	Native	Wrapped
Rebasing	• **Origin OUSD** USDC collateral earns income through lending and liquidity provision. Yield increases eligible holders' OUSD balances automatically. One token targets US$1. Token balance grows. • **BlackRock BUIDL**: Targets US$1 per token; monthly dividends arrive as additional BUIDL tokens.	• **Ondo rUSDY** Holds USDY in a wrapper contract. As USDY's reference price rises, the holder receives more rUSDY, with each rUSDY targeting US$1. The wrapper turns price growth into balance growth. • **Ondo rOUSG**: Wraps OUSG; the token balance grows. • **Aave aUSDC**: USDC lending receipt; balance grows.
Accumulating	• **Ondo USDY** A tokenised note secured by Treasury and deposit assets. Yield increases its redemption value per token; the holder's token count stays unchanged. One token represents a growing dollar value. • **Circle USYC**: Fund token; Treasury and reverse-repo income increases its token price. • **Ondo OUSG**: Fund income increases value per token.	• **Origin wOUSD** Wraps rebasing OUSD. The wOUSD balance stays fixed; each token unwraps into more OUSD as yield accrues. • **Ethena sUSDe** Deposit USDe to receive sUSDe; rewards increase the USDe recoverable per token. • **Sky sUSDS**: Deposit USDS; each sUSDS becomes redeemable for more USDS as savings accrue.

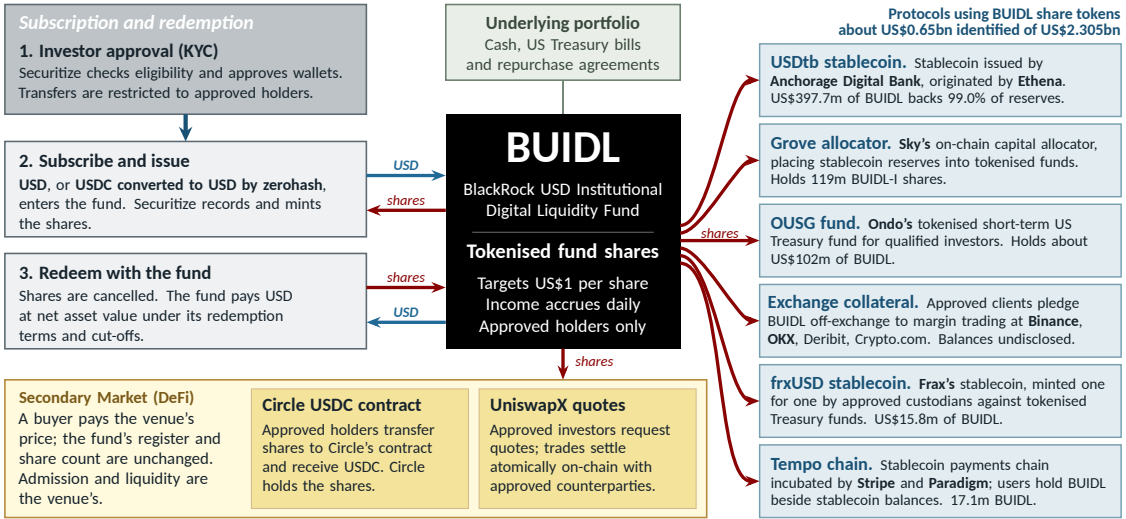
Native: yield accrues in the base token. **Wrapped**: a separate token represents underlying tokens held in a contract. **Rebasing**: yield changes the token balance. **Accumulating**: yield changes redemption value per token. The underlying legal claim can be a note, fund interest or protocol token.

Tokenised cash-management instruments

Provider / platform	Role and instruments	Selected networks	Assets US$bn	Transfers 30d	Value US$m, 30d
Ondo	Investment platform: **USDY**, secured Treasury / deposit note; **OUSG**, limited-partnership interests in money-market funds / short-term assets.	Ethereum, Stellar, Solana, XRP Ledger	**2.250** **0.396**	82,191 44	205.3 116.9
Circle	Token administrator for **USYC**: shares in Hashnote's liquidity fund, invested in Treasury bills and reverse repo; converts to USDC.	Ethereum, Solana, Canton	**2.506**	55	809.2
Benji (Franklin)	Franklin Templeton's fund-register platform: **iBENJI**, BVI institutional liquidity fund; **BENJI (FOBXX)**, US government money-market fund.	Ethereum, BNB Chain, Stellar, Solana, Aptos	**1.705** **0.672**	38 26	30.6 18.2
Securitize	Tokenisation and transfer-agent platform for **BlackRock**'s **BUIDL** USD liquidity fund; maintains investor records and supports share transfers.	Ethereum, Solana, Aptos	**2.305**	164	962.7
WisdomTree	Asset manager and fund platform: **WTGXX**, a US Treasury money-market fund, available through WisdomTree Prime and Connect.	Ethereum, Stellar, Solana	**1.230**	13	0.1
JP Morgan	J.P. Morgan Asset Management: **JLTXX**, US government money-market fund; **MONY**, private Treasury money-market fund.	Ethereum	**0.801** **0.103**	6 0	50.7 0
Libeara	Standard Chartered-backed tokenisation platform: **ChinaAMC**'s **CUMIU** digital money-market fund, Class I; **Bosera**'s **BELIF** liquidity income fund.	Ethereum	**0.554** **0.103**	0 0	0 0
Superstate	Tokenisation and transfer agent for **USTB**, **Invesco**'s short-duration Treasury fund; subscriptions, redemptions and transfers.	Ethereum, Solana, Plume	**0.571**	1,973	189.1
Centrifuge	Fund-tokenisation protocol: **JTRSY**, the **Janus Henderson / Anemoy** Treasury fund; tokenised ownership, subscriptions and redemptions.	Ethereum, Base, Stellar	**0.386**	73	1,513.4
OpenEden	Treasury-tokenisation platform: **TBILL** represents shares in a BVI liquidity fund; investors subscribe and redeem through a contract that handles purchases and redemptions of fund tokens.	Ethereum, XRP Ledger	**0.296**	23	28.6
Galaxy	Tokenisation infrastructure for **State Street**'s **SWEEP** private liquidity fund; PYUSD subscriptions and redemptions.	Solana	**0.158**	1	9.3
Spiko	Cash-management and tokenisation platform: **USTBL**, a French UCITS Treasury fund; subscriptions, ownership records and token transfers.	Ethereum, Starknet, Stellar	**0.154**	373	134.5

USDY: secured note; other instruments: fund / limited-partnership interests. BVI: British Virgin Islands; UCITS: Undertakings for Collective Investment in Transferable Securities.

BUIDL: *par, secured, yield-bearing*



BlackRock manages investments · **BNY** holds assets and administers the fund · **Securitize** maintains the share register · **zerohash** converts USDC subscriptions to USD.

Deposit token design space

	Single-bank arrangement	Multi-bank arrangement
Private ledgers	• **Kinexys Digital Payments**: J.P. Morgan *(authoritative deposit ledger)* • **HSBC TDS** *(mirrors of deposits)* • **Citi Token Services** *(Cash: interbranch liabilities)* • **DBS Token Services** *(tokenised deposit balances)* • **BNY tokenised deposits** *(mirrors of deposits)* • **Standard Chartered tokenised deposits** *(tokenised deposit balances)*	• **Partior**: participating settlement banks issue • **DeCurret DCP / DCJPY**: multi-bank design; one live issuer • **Great British Tokenised Deposit** *(Quant software coordinates payments between bank systems; live pilots 2026)* • **Hangang** *(real-value pilot)* • **Cari Network** • **Project Keystone**
Public ledgers	• **JPM Coin (JPMD), on Base** J.P. Morgan deposit liability on a public Ethereum layer-2 network; approved institutional holders. • **HSBC TDS on Canton** Simulated issuance and atomic settlement; April 2026 pilot.	• **PostFinance / Sygnum / UBS** *Swiss Deposit Token proof of concept (2025):* public chain, permissioned apps; payment instructions; cash settlement through SIC. • **CBMT public-chain route** *(testnet only)* Public testnet in the sandbox architecture, alongside private Besu chains and inter-ledger bridges.

Operator services; public chains can restrict holders. Production status and legal claims are separate. TDS: Tokenised Deposit Service; DCP: DeCurret platform; DCJPY: yen deposit token; CBMT: Commercial Bank Money Token; SIC: Swiss Interbank Clearing.

Deposit token networks

Pilots & development

Arrangement	Issuer / claim	Interbank settlement	Technology	Status / evidence
Swift ledger 17 pilot banks	Each bank issues its own deposit liabilities.	**Central-bank money** (RTGS) or **commercial-bank money** (correspondents).	Besu	Live transactions from Aug 2026; controlled pilot.
DeCurret DCP 43 pilot firms	DCJPY: yen deposit token. Issued by GMO Aozora Net Bank.	**Commercial-bank money** (lead bank) or **stablecoins** in interbank settlement tests.	DCP: DeCurret platform; Besu	Aggregate unavailable Payment value
CBMT 7 sandbox banks	Each bank issues its own test-money token.	**Central-bank money** envisaged for settling interbank balances.	Besu	Sandbox; test currency only.
Hangang 9 Phase 2 banks	Bank tokens; wholesale CBDC.	**Central-bank money:** Bank of Korea wCBDC settles between pilot banks.	Besu	Phase 1 completed; Phase 2 authorised Jul 2026.
GBTD 7 pilot institutions	Each institution issues its own deposit liabilities.	**Central-bank money:** atomic settlement explored in the BoE Synchronisation Lab.	Quant platform[1]	Customer pilots completed Sep 2026; asset pilots next.
Cari Network 30+ joining / 6 design	Banks would issue their own deposit tokens.	**Commercial-bank settlement model:** daily net fiat settlement via a designated bank.	ZKsync Prividium (Ethereum L2)	Bank onboarding in preparation.
EnsembleTX 7 pilot banks	Banks issue tokenised deposits.	**Central-bank money:** HKD RTGS initially; tokenised central-bank money planned.	Bank ledgers[2]	Real-value pilot through 2026.
e-CNY 2.0 2026 framework	Bank-wallet balances are commercial-bank deposits.	**Central-bank money:** PBoC settlement wallets funded with bank reserves.	Accounts + DLT	Deposit regime from Jan 2026; ongoing pilot expansion.
LSEG DiSH Bank testing	Ownership of commercial-bank deposits, recorded on DiSH.	**Commercial-bank money:** deposit ownership transfers recorded on DiSH.	Canton	Jan 2026 technical launch; repo PoCs; 2027 Partior link.
Project Keystone 5 named / 6 reported	Bank-administered claims on regulated deposits.	**Not specified:** central- or commercial-bank money for interbank settlement.	Not announced	In development; no live volume reported.
The Clearing House 17 backing banks	Banks would issue their own deposit tokens.	**Central-bank money** via planned RTP / CHIPS links; on-chain asset not yet specified.	Not announced	Planned for H1 2027.

See Industry Ledgers Table 1 for mature deposit token projects. GBTD: Great British Tokenised Deposit; CBMT: Commercial Bank Money Token; DiSH: Digital Settlement House; LSEG: London Stock Exchange Group; MTN: Multi-Token Network.

Par reachability for money tokens

"Everyone can create money; the problem is to get it accepted." – Hyman Minsky

Par for money tokens means **one-for-one convertibility into central bank money**, the terminal settlement asset. RITS anchors par across Australian banks by settling interbank obligations through **Exchange Settlement Accounts** of reserve balances. The introduction of new types of par value tokens based on distinct legal arrangements complicates monetary singleness. Tokens transferred in payment must be accepted by the recipient as par-redeemable.

Tokens that transfer value through a **burn-mint mechanism** must be funded through an inter-issuer arrangement; this could settle in central bank money (equivalent to ESA balances in RITS) or commercial bank money, the latter subject to settlement risk. Fees, delays, and haircuts along the conversion path can weaken effective par and the singleness of money.



1. **Access.** The holder may lack an issuer or intermediary willing to redeem the token at par.
2. **Hours.** RITS reservations end in the evening. The FSS runs 24/7 without reservation.
3. **Capacity.** Limits on redemption throughput: intraday liquidity, batch cycles, agent bank exposure limits.
4. **Credit.** Par exchange links claims on different issuers, with different credit risks.

Ubyx "cheque" clearing model. Bank or fintech presents tokens to the issuer for redemption, then credits the customer after settlement. A token can be both a transferable issuer claim and a payment instrument. Unpaid tokens are returned; the collector does not guarantee payment. Multilateral interchangeability between different issuers' money tokens requires a multi-issuer acceptance network or **scheme**.

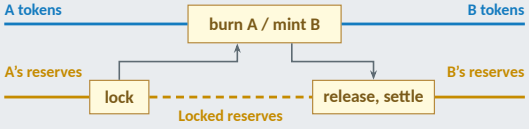
Par token interchange models

One issuer's tokens must be exchangeable for another's in the same currency at face value.

1. Sequential exchange

Burn A's tokens, transfer reserves through RITS, then mint B's tokens for the recipient.
Separate steps create timing and recovery risks.
Example: RBA simple exchange model.



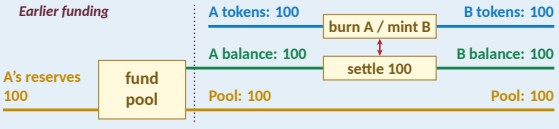


2. Reserves-lock exchange

Lock A's reserves in RITS before the token actions.
Release them to B once those actions complete.
Locks reduce settlement risk. Burn and mint must be atomic.
Example: proposed RITS reserve-lock model.

3. Prefunded utility: gross settlement

Prefund A's utility balance with pooled reserves. Each payment debits A and credits B, with customer tokens burned and minted.
Exchange can be sequential or lock A's reserve-backed balance.
24/7 settlement requires prefunded liquidity.
Settlement analogue: RTP.



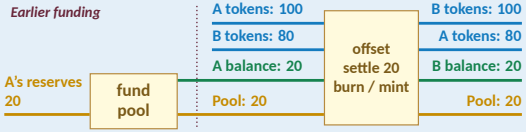
Par token interchange models

One issuer's tokens must be exchangeable for another's in the same currency at face value.

4. Backed exchange token

Prefund segregated reserves and issue E to A.
Burn A tokens, transfer E to B, then mint B tokens.
Backing stays in its reserve account, tying up liquidity.
Settlement-asset analogue: Fnality.





5. Netting: either arrangement

Queue and offset payments: A pays B 100; B pays A 80.
Both token payments complete in full; settle the net 20
using utility balances (3) or exchange tokens (4).
Netting saves liquidity; queued payments are final on release.
Settlement analogue: CHIPS.

6. Deferred interbank settlement

Exchange A tokens for B tokens and record A's debt to B.
Settle the interbank debt later through RITS, gross or net.
Payments can continue outside RTGS hours; banks carry
credit exposure until interbank settlement.
Example: Swift ledger model described by the RBA.



At-par exchange: RITS requirements

- **Membership and holder rights.** Confine par interchange to a **common instrument type** (deposit tokens). Publish risk-based issuer licensing and access criteria; specify holder rights, segregated backing, and direct ESA access or a sponsor.
- **Operator rules and redemption.** Define authority to *reserve*, *sequence*, *route* or *net* obligations. Set infrastructure supervision, finality, loss allocation and recovery rules. Standardise *hold*, *burn*, *mint*, *settlement*, *cancellation* and *status* instructions, with coordinated completion or cancellation. Provide continuous redemption, with liquidity and capacity matched to token use.
- **Settlement-token backing.** Distinguish an RBA liability, a private ESA-backed claim and a private collateralised claim. Specify the issuer, operator, insolvency treatment, collateral eligibility, haircuts, limits and loss absorption.
- **RBA policy on tokenisation and AUD settlement.** Set policy on private money, foreign-currency domestic settlement, offshore routing and open-mint liquidity. Assess the instruments made accessible and vendor concentration in custody, wallets and key management.

CPMI-IOSCO, 2022, on stablecoin arrangements: a settlement stablecoin should give holders a clear legal claim on the issuer or its reserves, be convertible at par into central bank money or other liquid assets at the latest by end of day from high-quality, segregated reserves, and have a defined, legally robust point of finality; probabilistic consensus alone is insufficient.

AREA 3

Reserve access

Q9. How might stablecoin issuers use access to central bank reserves if it was made available, both today and as arrangements scale?

Q10. What account features would be most valuable to stablecoin arrangements, both today and as they scale?

Q11. What barriers, other than eligibility, would stablecoin issuers face in accessing and using central bank reserves?

Backing, access, and remuneration

- **Reserves as backing.** Bank-deposit backing exposes the reserve holder to bank credit risk; the token holder's claim depends on the issuer's terms. Bank failure can impair recovery or delay access. Government securities carry price and liquidity risk. **Central bank money avoids commercial-bank credit risk**; redemption still needs operational access and liquidity arrangements.

- **Forte Stablecoin (Project Acacia).** An ADI held matching ESA balances under a private contract with Forte to back its **AUDF** stablecoin. Those reserves were not legally or operationally segregated from the ADI's other ESA holdings; coin holders had no claim on the RBA. ***A direct account changes the issuer's counterparty from an ADI to the RBA.***

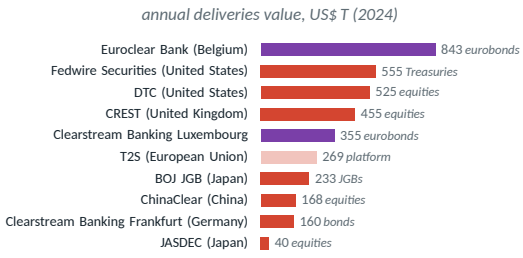
- **Account models for consultation.** A full **ESA**; non-ADI stablecoin issuers are generally ineligible. A **settlement-only** account, as in the Fed proposal, capped and unremunerated. A **reserve-only** account transacting through an agent. An **omnibus** account on the Bank of England model, where the operator rather than the issuer holds it; CHIPS has settled through one at the Federal Reserve since 2001.

- **Remuneration.** Reserve income is a major source of issuer revenue. The Bank of England leaves stablecoin backing reserves unremunerated, treating stablecoins as payment instruments rather than savings products.

Reserve account types

The *Principles for Financial Market Infrastructures (PFMI)* permit other settlement assets where central bank money is not practical or available, subject to **strict control of credit and liquidity risk**. Controls include settlement-bank creditworthiness, exposure limits, final transfers and access to proceeds by the end of the day.

- **Exchange Settlement Account (ESA).** Normal reserves account able to support backing, payments and interchange.
- **Settlement-only account.** Payments only. Fed proposal: capped, unremunerated, no credit.
- **Omnibus account.** Pooled backing; the UK model requires a recognised, supervised payment operator.
- **No account.** Commercial bank money: industry accepts settlement risk, and a bank token (e.g. JPM Coin) can be the settlement asset.



Euroclear Bank and **Clearstream Banking Luxembourg** are the two *International central securities depositories* (ICSDs): depositories originally set up to settle Eurobond trades, that now settle internationally traded securities across currency areas. *Eurobonds* are issued, and largely sold, outside the domestic market of the currency in which they are denominated.

The ICSDs are among the largest financial market infrastructures in the world, holding EUR42.5 trillion and EUR21 trillion of assets under custody. Both are licensed banks that settle the cash leg in their own **commercial bank money**: the largest standing exception to the principle of settlement in central bank money.

Omnibus central bank reserve accounts

System	Legal entitlement	Settlement
Fnality £FnPS	**Trust / beneficial ownership.** The operator holds the BoE account on trust for participants; their interests in the reserve pool are recorded on the ledger. Enabled by the BoE's April 2021 omnibus-account policy; balances are backed 1:1 by central bank funds.	DLT transfers; cash can be earmarked for delivery-versus-payment / payment-versus-payment. Live since December 2023, with statutory finality designation in December 2024.
CHIPS	**Joint account / scheme obligations.** Funding banks jointly hold the Fed account. CHIPS balances are joint obligations of those banks, payable only from the pool. CHIPS does not own the funds.	Queuing and offsetting save liquidity; payments are final on release. Fedwire funds the operating day and closing payouts.
RTP network	**Joint-benefit prefunding.** Funding banks / agents share the pool; TCH controls payouts as agent. An individual RTP position is not a separate claim on the Fed.	Gross settlement, 24/7, against prefunded positions. Customers continue to hold deposits at their banks.
EURO1 (EBA Clearing)	**Single obligation structure.** Each participant has one claim on, or obligation to, all other participants as joint creditors and obligors; all participants contribute equal shares to a liquidity pool of cash deposits held with the ECB.	Offsetting through the day; net settlement at end of day in T2. The pooled deposits are held at the ECB only during operating hours, returning to each participant until the next T2 day, where the CHIPS pool stands overnight.
RT1 (EBA Clearing)	**Technical account; operator as administrator.** RT1 holds a TIPS technical account at the ECB “in an administrative capacity only”. The pool records Participants’ positions.	Instant euro payments settle in central bank funds against prefunded positions, 24/7.
Alipay / WeChat Pay wallets	**Segregated statutory pool.** Since January 2019, 100% of customer funds sit at the PBoC, ring-fenced from the institution's creditors; wallet holders' claims are on Alipay or Tenpay, not the PBoC.	NetsUnion / UnionPay clear wallet payments; the PBoC pool funds settlement. Historical rate: 0.35% from Aug 2019; not a current-rate claim.

Each scheme has its own legal structure; full backing alone does not establish who owns the reserve balance or who owes the holder. Omnibus accounts also exist for commercial bank money settlement, for example the **Cari network**.

Tokenised repo and CoBM settlement

Tokenisation may introduce new types of reserve-holder while encouraging some existing reserve-holders to shift toward settling with *claims* on reserves (i.e. deposits). Central banks use repurchase agreements (repos) to control the supply of liquidity and influence short-term interest rates. For AUD repo conventionally settled through Austraclear, **the interbank cash payment settles in central bank money (CeBM) in RITS.**
Accepting another bank's deposit tokens settles the interbank obligation in **commercial bank money (CoBM).**
Access to and use of central bank money was explored during the RBA's Project Acacia, where one use case brought together **CBA**, Australia's largest bank; **JPMorgan**, the world's largest bank by market value, and its **Kinexys** tokenisation platform; **ASX**, operator of **Austraclear**, Australia's domestic securities depository for government debt; and **HQLAx**, a blockchain platform that transfers collateral ownership while securities remain with their custodian.

Project Acacia intraday repo



***Intraday repo**: cash lent against collateral and returned within the same day. ESA holders today obtain it from the RBA's interest-free intraday repo in RITS (unwound by end of day). Tokenised intraday repo lets banks obtain it from each other.*

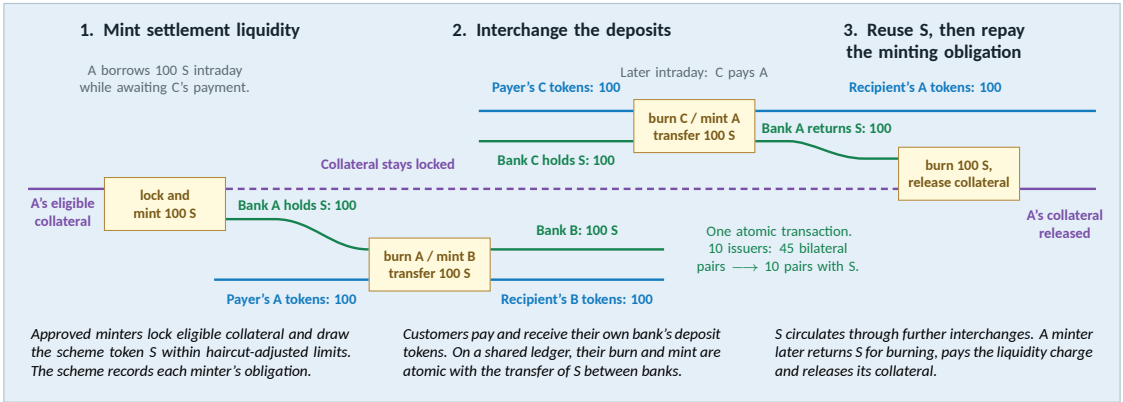
Settlement in commercial bank money. Kinexys settles with commercial bank money, not central bank money like Austraclear. CBA reportedly also demonstrated a "simulated proof of concept" where simulated "wCBDC" was used to settle an interbank obligation, but was not involved in Acacia's live pilot wCBDC experiments. The RBA has heretofore required central bank money settlement of Australian repo, per PFMI standards.

A Scheme Settlement Stablecoin

On-demand settlement liquidity against eligible (programmable) collateral

Programmability could enable a new type of reserve access: approved participants mint a common token on demand against RBA-repo eligible collateral. These privately issued balances act as "reserves" for multilateral deposit-token interchange.

An open-mint facility would apply the scheme's collateral rules (haircuts and issuance limits) automatically. Token liquidity could be priced by the amount drawn, the time outstanding, and the collateral risk. Returning the settlement token for burning retires the issuance obligation and releases the backing.



Reserve access: RITS requirements

- **Eligibility and holder protection.** Specify conditions under which licensed issuers may hold reserves, through which account and with what settlement rights.
- **Finality and limits.** State which transfers gain *PSNA* finality, whether accounts are capped, and whether non-ADI issuers will have any access to intraday credit.
- **Cash access and liquidity.** Tokens circulate continuously. Support continuous funding and redemption with issuance matched to funding; set redemption buffers, stress funding arrangements and any RBA facility access, with eligible collateral, haircuts and limits.
- **Remuneration.** Interest supports the commercial viability of full reserve backing, especially where it cannot be passed to holders. However, the central bank typically pays interest on reserves that are not costless to acquire, at rates set according to monetary policy objectives. Remuneration of reserves held against RBDC issuance would represent an effective public subsidy for narrow banking.
- **Reconciliation and reporting.** For any fully reserve-backed token, reconcile supply with segregated backing, including in-flight issuance and redemption.

UK June 2026 policy / draft rules: at least 30% unremunerated BoE reserves; up to 70% permitted short-term government assets. Liquidity backstop planned.

AREA 4

Tokenised reserves

Q12. Which use cases would become materially easier or more efficient if tokenised reserves were available?

Q13. What functional outcomes should the design of tokenised reserves, and the issuance and distribution model need to support?

Q14. Based on the use cases that require tokenised reserves and the outcomes that need to be supported, what issuance and distribution approaches would be appropriate?

Q15. What funding, redemption and liquidity-management capabilities would be required between ESAs and tokenised reserve holdings?

New forms of central bank money

- **Central bank money off the platform.** Consultation areas 1 to 3 assume RBA central bank money stays in RITS, reaching external platforms through synchronisation or reserve access. Synchronisation references it from the platform with a hold; exchange reaches it through an interbank payment; reserve access changes the backing asset. Tokenised reserves (Area 4) place central bank money on new platforms external to RITS.

- **Tokenised central bank money (CeBM).** A **digital twin** represents a segregated ESA balance, with token supply kept aligned to that balance (Pontes design). T2 funds are escrowed in a **Token Issuance Account** under ECB legal responsibility. Matching cash tokens are claims on the ECB to transfer funds in T2; redemption returns funds to participants' T2 accounts. A native wCBDC records a direct central-bank claim on the token ledger. Issuance may still debit or segregate existing reserve balances.

- **Ledger operator.** If the RBA operates the ledger, external platforms synchronise to it as in Area 1 (this is the **Pontes** option chosen by the ECB). If a third party operates it, an RBA liability settles on infrastructure the RBA does not run (this is the **Project Helvetia** option chosen by the Swiss National Bank, which has minted wholesale CBDC on a ledger operated by SIX Digital Exchange).

- **Hours of operation.** A token can circulate continuously; conversion and liquidity between token and reserve balances will be constrained when issuance and redemption can only occur during business hours. Pontes will initially settle only within T2's operating day; continuous operation is planned for its 2028 version.

Tokenised CeBM

Design choices

Form of the token — digital twin | native mint

a twin mirrors a separate authoritative balance; a native token ledger records the central-bank claim itself

Ledger operator — RBA-operated | third-party

cash-ledger operation and oversight; same-ledger atomicity also needs assets and suitable transaction logic

Distribution — single tier | two tier

single tier: only ESA holders hold the token, for themselves; two tier: they also hold it for clients

Issuance hours — business hours | continuous

continuous circulation with limited conversion hours constrains liquidity; it need not cause a price discount

Atomicity vs control

	Liquidity pools to fund	Cash-ledger operation	Atomicity with external assets
Issued onto many platforms	one per platform, plus RITS	multiple external operators	possible, platform by platform
Issued onto one external platform	two	one external operator	possible on that platform
Issued onto RBA-operated platform	two	RBA-operated	needs cross-ledger synchronisation
No tokenised reserves issued	one	RBA-operated	needs cross-ledger synchronisation

The RBA can retain mint/burn and eligibility controls on external ledgers, as tested in Acacia. Atomic execution on one ledger requires both assets present. Cross-ledger synchronisation can also provide all-or-nothing settlement, as designed in Pontes.

21 September 2026
settles within T2's day; finality in T2

→

planned enhancement
22.5h per business day;
finality on the Eurosystem DLT

→

target: mid-2028
24/7, multi-currency and greater programmability

Pontes launched with Hash-Link and cash settlement in T2. Its planned DLT cash-token will mirror dedicated T2 reserves, leaving final cash settlement with RTGS instead of introducing a true wCBDC. For native reserves, the token ledger is the authoritative record; legal finality follows the applicable framework.

Design space for tokenised reserves

	Cash-token ledger operated by central bank(s)	Cash-token ledger operated externally
CBDC	• **mBridge** *(mBridge Ledger)*: shared central-bank platform; minimum viable product reached in 2024. • **e₹-W**: Indian wholesale pilot, since 2022. • **Hangang**: Korean central-bank-money pilot component, used as settlement asset for deposit token interchange. • **SGD Testnet**: MAS-operated shared ledger; first live SGD wholesale CBDC issuance, 2025.	• **Helvetia** *(SDX, Corda)*: Swiss National Bank sight deposits in token form on SIX; pilot extended to at least June 2028. *SIC, the Swiss franc RTGS, is run for the SNB by bank-owned SIX. Hong Kong (HKICL) and Canada (Payments Canada) also use non-central-bank operators.*
Digital Twin	• **Pontes cash-token design** *(DL3S, Hyperledger Fabric)* Tokenised central bank money (CeBM) on an ECB-operated DLT platform, representing funds in a T2 account. Cash finality remains in T2.	• **Acacia** *(Canvas Connect, Ethereum, Hedera, Redbelly, XRP Ledger)*: RBA pilot wCBDC tokens, 2026. *Although intended to model a true wCBDC, the pilot tokens were minted against conventional ESA balances that were held in reserve for the duration of their issuance.*

Synchronised RTGS settlement for interchange with no central-bank token:
initial stage EnsembleTX / HKD RTGS; UK RTGS synchronisation; SeturionX / SIC.

A multi-asset *RITS-DLT* platform?

- **Assets on the money ledger.** Europe's **Pontes DLT** (not yet live) will be able to synchronise with external platforms (via Hash Link), but will also be able to host tokenised assets *internally* and settle with *same-ledger* atomicity. This is effectively a tokenised version of **T2S**, which brought securities accounts from European depositories into the same system as cash accounts from the central banks (depositories keeping the register and legal responsibility). T2S settles securities atomically with central bank money on its own Eurosystem-owned and operated platform.
- **RITS began as a securities settlement system.** All four RITS Consultation areas assume that tokenised assets will sit externally to RITS. However, **RITS settled Commonwealth Government securities** from 1991 to 2002, before the RBA transferred that role to **Austraclear**. Other central banks serve as CSDs for sovereign debt, including the US Federal Reserve and the Bank of Japan.
- **Netting versus atomic settlement.** Where central clearing is used, trades can be netted before settlement. T2S settles each instruction gross against gross funds (**delivery-versus-payment Model 1**); CHESS settles net securities against net cash in a daily batch (**Model 3**). Austraclear uses Model 1. Atomic per-trade settlement reduces netting opportunities for these trades.
- **Gross settlement needs a liquidity mechanism.** DvP Model 1 requires liquidity on demand. T2S has **auto-collateralisation**, firing inside the settlement transaction against the securities being bought or eligible debt held. The RBA's interest-free intraday liquidity facility is member-initiated and closes with Austraclear at 6:30pm.

Project Atom, 2020–21, was an RBA test of a multi-asset ledger arrangement. A tokenised wCBDC funded, settled, and repaid a tokenised syndicated loan on an Ethereum-based platform, the CBDC and the asset on one ledger, and settlement across platforms was tested with token holds and hashed time-locked contracts.

An Orchestration Layer

A proposal to separate **core RTGS** requirements – high availability, low latency and final settlement – from a **modular overlay service**.

Liquidity Savings Mechanisms (LSMs)

LSMs optimise payment order before submission to RTGS. Most RTGS systems have queue-management systems able to offset payments multilaterally; the RITS LSM is only able to offset bilaterally.

Netting vs resequencing

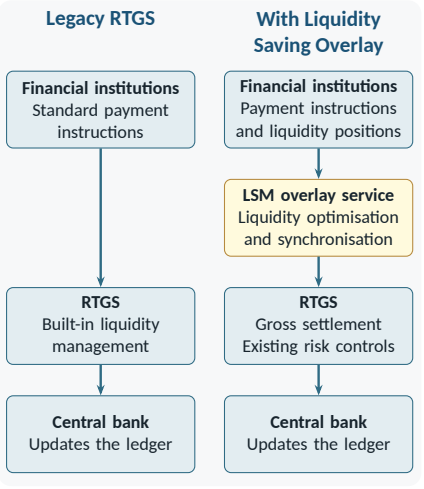
- **Netting** changes legal obligations by offsetting them, leaving only net amounts payable.
- **Resequencing** changes order, preserving every original gross obligation. All payments must still settle; the first still needs funding. Payments submitted to clear on a gross basis cannot legally be netted and batch-settled, but can be held for resequencing.

Computational difficulty

Liquidity-optimal resequencing is NP-hard in general, but polynomial-time solvable under centralised routing. Existing RTGS LSMs optimise over offset selection, not sequence selection.

Quantum optimisation of payment sequences

Hybrid quantum–classical preprocessing cut simulated liquidity needs (Bank of Canada; Bank of Italy).

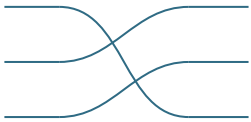


Fedwire runs pure RTGS with daylight credit and no LSM. **Gridlock resolution** covers circle search (**SIC**), bilateral offset (**RITS** auto-offset and targeted bilateral offset) and multilateral offset (**Kronos2**, **Lynx**, **CHAPS**, **BOJ-NET** and **HKD CHATS**); **T2** dissolves its queues with optimisation algorithms aimed at the highest settlement volume for the least liquidity. **CHIPS** uses stable matching, a patented (now expired) balanced-release algorithm trading liquidity savings against settlement delay.

Resequence, net, route

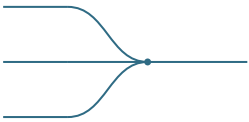
Where wholesale payments can settle or wait, **resequencing** allows substantial liquidity savings. RITS's queue offers **priority statuses** and **bilateral auto-offset** but no **multilateral offsetting algorithm** of the kind T2 (Eurosystem), CHAPS (UK), and Lynx (Canada) run. An orchestration layer that sequences an Australian dollar obligation, Swift's shared ledger, Canton's Global Synchronizer, or a central synchronisation operator for multilateral exchange, would also choose whether the obligation finally settles in RITS, or in another system — and possibly another currency.

Resequencing



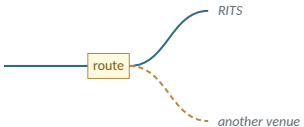
same obligations, different order, legal form unchanged

Netting



one obligation replaces three, legal form changes

Routing



same obligation, different venue; the place of finality changes

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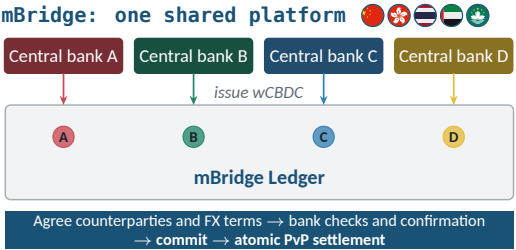
RITS settles gross and real-time, but also runs multilateral group settlement for batch and retail payment systems. Once cross-border ledgers are interoperating, domestic AUD obligations between two Australian banks could potentially be routed over an orchestration layer to settle in external multi-currency systems that the RBA does not govern. As industry shifts to a tokenised financial system that operates 24/7 across global markets, the RBA must establish whether RITS will retain its exclusive legal designation for settlement finality over domestic interbank payments.

Project Agorá

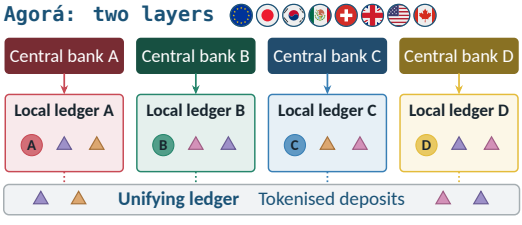
Correspondent banking routes cross-border payments through chains of **nostro accounts**, resulting in delays, limited traceability, and prefunding requirements at each intermediary. These frictions contributed to the rapid adoption of payment stablecoins, including for card scheme settlement. The **Bank for International Settlements** and various central banks have undertaken research projects to test the use of tokenised central bank money to improve cross-currency settlement.

One of these projects, **Project mBridge** (2021–present) advanced to a **Minimum Viable Product** stage in 2024.

The mBridge architecture leaves no role for correspondent banking in cross-border PvP, with settlement made in central bank money on a ledger facilitating direct settlement in CBDC. Project Agorá, launched in 2024, retains correspondent banking arrangements: each central bank's money remains on its domestic ledger, while commercial bank money circulates on a shared international ledger.



A buyer in A pays a seller in B. *The buyer's bank debits its customer's deposit and sends A's CBDC to the seller's bank, which credits the seller's deposit. The receiving bank can hold, reuse, or exchange that CBDC. If the seller wants B's currency, a bank supplies FX: the two CBDCs can be exchanged atomically through PvP.*



A buyer in A pays a seller in B. *Banks and correspondents agree the route and FX amounts, validate the payment, and lock balances. Customer and correspondent deposits change on the unifying ledger; reserves move on domestic ledgers. All legs follow one commit-or-cancel decision. The seller receives a deposit in the agreed currency.*

● Tokenised CeBM/CBDC ▲ Tokenised deposit (CoBM)

Agorá: Jurisdictional Ledger requirements

The Agorá report sets out fifteen business requirements. Each leaves scope for a variety of implementations.

1. **Jurisdictional autonomy and governance.** Central banks retain control of reserves, monetary policy, participation, and domestic operations.
2. **Issuance, redemption, and transfer.** Preserve legal claims and balance sheets, with auditable movements and exceptional issuer controls.
3. **Atomic settlement of every leg.** All legs succeed or fail together, across currencies and ledgers.
4. **Pathfinding and identification.** Agree eligible routes and verify correspondent links, respecting each bank's routing rules.
5. **Cross-currency integrity.** Apply agreed currency amounts consistently across every payment leg.
6. **Access, participation, and authorisation.** Define token access and limit delegated actions; originating participants remain accountable.
7. **Compliance pre-validation and screening.** Settle only after each bank's sanctions, AML/CFT, fraud, and balance checks pass.
8. **ISO 20022 data standards, quality, and interoperability.** Preserve payment data across systems; validate identifiers, amounts, and required fields.
9. **Reduction of exceptions and investigations.** Use early checks, clear responsibilities, timeouts, and unambiguous cancellation outcomes.
10. **Transparency, notifications, and reporting.** Give authorised participants timely status, confirmations, balances, and audit trails.
11. **Privacy and selective data-sharing.** Share only necessary data with authorised participants, within local rules.
12. **Liquidity management and monitoring.** Track reserve and deposit balances across currencies, 24/7.
13. **Governance and allocation of roles and responsibilities.** Institutions retain customer servicing, compliance, and reconciliation duties.
14. **Platform reliability, security, and scalability.** Meet measurable performance and availability standards, with predictable recovery from outages.
15. **A clearly defined settlement-finality trigger.** Define a legally enforceable finality point across jurisdictions, including in insolvency.

The Swift Ledger

Swift is the messaging network underpinning correspondent banking. Direct cross-border CBDC settlement would bypass this network, but Agorá preserves its role, positioning Swift as a natural vendor to supply its unified ledger.

Swift announced its ledger in September 2025, during Agorá's prototype development. Both use **Hyperledger Besu**. Swift calls it a "shared digital orchestration layer": coordinating payment commitments while *final* settlement uses existing systems.

Tokenised reserves: RITS requirements

- **wCBDC vs "tokenised CeBM".** A digital twin can represent an ESA balance. That balance must sit in an account that cannot be spent through RITS while tokens are outstanding. Every mint or burn must be matched by a transfer into or out of it, so token supply and balance stay equal. A native token (wholesale CBDC) would be a direct RBA liability, and require its own legal basis for finality, as the RITS Regulations give ESA movements today.
- **Agorá compatibility.** A jurisdictional reserve ledger under the RBA's own authority, carrying tokenised ESA balances rather than a wCBDC, with locking, cross-domain coordination, embedded compliance, and finality under Australian law.
- **Public or private ledger operator.** An RBA-run ledger means every external platform still synchronises to it; a third-party ledger requires enforceable observation, access, governance and recovery rights, as well as clear issuer control. Any private operator would need to be domestic, credibly neutral, and user-owned; AP+ or a national industry consortium of a similar type.
- **Hours of operation.** Continuous circulation against business-hours issuance and redemption opens a conversion constraint between two RBA liabilities. The Pontes plan for continuous operation in 2028 requires tokens to remain outstanding outside T2 hours, with finality moved from T2 onto the Eurosystem's own DLT platform.
- **Intraday credit on the ledger.** Settling every payment in full requires each bank to hold its peak gross position on the ledger. RITS meets this need by intraday repo during business hours. A 24/7 ledger warrants a similar type of facility, with programmable money and assets broadening the range of possible mechanism designs.

Submission responses

which do **not** represent the positions or policies of any organisation

Area 1: Synchronisation

Q1 · Synchronisation models

RITS should be capable of supporting central bank **reserves lock 24/7**. While RITS cannot lock reserves on an external platform it doesn't control, it should be able to interface with external platforms through a **ISO20022-based API** that supports DvP with asset-locking on those platforms. The API should offer **asset-lock, reserves-lock or dual-lock DvP/PvP** support, accommodating counterparty preferences for any potential asset transaction (e.g. real-estate, securities, FX) settled under a future national payments scheme.

FSS is the prefunded retail support that operates during RITS "after hours", without support for reservation locking. FSS could be upgraded to support locks, DvP, synchronization, and wholesale payments; in doing so, the enhanced FSS would effectively become the **main settlement engine**; so RITS would operate 24/7 'normally'.

Q2 · Synchronisation arrangements

The RBA should provide an interface to central bank reserves for third-party operators including **Australian Payments Plus** as user-owned operator of an interbank AUD settlement facility synchronizing RITS with programmable asset platforms. The utility would be governed by an industry scheme under regulatory oversight, guaranteeing **PSNA finality** on both payment and asset side. The utility should support AUD deposit token interchange and DvP/PvP settlement, including **T2S-style auto-collateralization**, using ISO20022.

This interface would be consistent with the requirements of the **BIS Agora framework** for maintaining a two-tier international monetary system.

Area 1: Synchronisation *(continued)*

Q3 · Existing RITS and FSS capabilities

RITS supports asset-lock links (Austraclear, CHESS) and reserves-lock batches (PEXA, Sympli), but only during business hours. FSS settles continuously, but has no reservation service. RITS feeder systems lack a common API, and there is no service for **credit-within-settlement** (which would enable **auto-collateralisation**). PayTo authorization cannot reserve ESA funds, and the NPP cannot support settlement synchronization in its current form.

An upgrade of the FSS to support reservation locks and 24/7 synchronized real-time settlement outside of the NPP will likely adversely affect the valuation of existing NPPA assets – but AP+ recognizes that the RBA will correctly disregard **existing payment infrastructures' sunk costs** when evaluating available options for RITS modernization.

Q4 · Required enhancements

The RBA should add **24/7 reservation functionality** and a **standardised ISO 20022 API** for settlement use cases, modeled on the Eurosystem T2S/TIPS. The RBA should clarify the extent of **RBA liquidity facility** support for continuous settlement in wholesale asset markets, and explore credit against purchased or pledged assets, including T2S-style **auto-collateralisation**.

This would support a new market in intraday liquidity, but would require integration between a modernized RITS and a modernized ASX Austraclear (or equivalent).

Area 2: At-par exchange

Q5 · Exchange models

RITS/FSS should support a national deposit token network with a 24/7 **multilateral mint–burn facility** synchronized to reserved ESA funds in RITS. The lock would reserve issuer A's payment to B while A's token is burned and B's token is minted; the reservation would then be released to settle.

A scheme facility would limit par interchange to a common instrument type: **deposit tokens**. Because stablecoins maintain par under different legal arrangements and without access to RBA liquidity, a RITS synchroniser should not be used to "peg" them to Basel-regulated deposits.

Q6 · Synchronisation arrangements for exchange

Exchange models should receive the same service for central bank supported interbank payment settlement as conventional account-to-account payments (i.e. HVCS, BECS or NPP), under the same types of rules. The RBA could provide the facility to the deposit token network directly (as in the Bank of Korea's **Project Hangang**), or a synchronizer could link an external facility (e.g. an AP+ token interchange) to RITS under the **accredited operator class**. Operators pre-processing payments for settlement would require infrastructure supervision, finality, loss allocation and recovery rules.

Area 2: At-par exchange *(continued)*

Q7 · Existing capabilities for exchange

The existing RITS/FSS could not support an effective reserve-collateralised **multilateral mint–burn facility** as RITS main operating hours are limited, the FSS lacks reservation functionality, and both lack an API.

Q8 · Required enhancements

Enhancements should include continuous reservation, redemption and – either directly or through an industry consortium, or operator – a common mint–burn service for AUD **deposit tokens**. Longer term, establish an API for a multi-issuer AUD intraday settlement token direct the formation of a **national industry scheme** for multi-issuer AUD intraday settlement.

ESA holders (and potentially non-holders) could mint and burn this token on demand against pooled tokenised HQLA, programmed according to the RBA's rulebook on **repo-eligible collateral**. This would improve access to secured intraday credit and contribute to AUD monetary singleness without requiring a new digital form of RBA liability or a reliance on a centralized mint-burn interchange.

Area 3: Reserve access

Q9 · Use of reserve access

Stablecoins should not be generally interchangeable for **deposit tokens** within a single token interchange facility, due to the substantive differences in the legal character of the instruments and their regulatory regimes. The RBA – through the **Deposit Token Working Group** – must clearly establish the role of AUD-denominated central bank money in deposit token interchange to differentiate them from stablecoins, both in wholesale and retail context – cf the role taken by the Bank of Korea in the **Project Hangang** deposit token network.

For stablecoins, a synchronization operator could lock ESA reserves for settlement and redemption, or for minting a fully reserves-collateralized token; an RBDC (or "**synthetic CBDC**"). Industry adoption of a fully reserves-collateralised stablecoin (RBDC) as a multilateral deposit token settlement asset would represent an effective privatisation of the RBA's core central banking function. Other forms of stablecoin may be minted or redeemed against eligible collateral backing held by trusted register according to an RBA rulebook.

Q10 · Account features

Issuers require support for continuous issuance, redemption, and settlement access. RITS/FSS reservation functionality with legal segregation and holder rights is a basic requirement for synchronised interchange. With these features, a scheme-governed open-mint AUD (or "**AUD+**") stablecoin could be maintained for on-demand tokenized settlement liquidity, based on an RBA rulebook and a trusted registry able to support locked par-value HQLA collateral.

Q11 · Barriers beyond eligibility

Unsegregated backing and liquidity constraints may obstruct stablecoin redemption into central bank reserves, causing their par to depeg. The RBA should develop a policy that would allow regulated stablecoin issuers to access a new, specialized **RBA liquidity facility** that could offer intraday credit for settlement finality against issuer's locked, **repo-eligible collateral** – c.f. the recent Bank of England proposal for regulated stablecoins.

Area 4: Tokenised reserves

Q12 · Use cases for tokenised reserves

Tokenised reserves refers either to wCBDC or to what the ECB describes as "**tokenized Central Bank Money**";technically not wCBDC because these tokens "wrap" ESA balances – being therefore the central bank liability's **digital twin** rather than the liability itself. Tokenised reserves would be created to settle transactions atomically on a common ledger with tokenized assets.

This ledger with tokenized reserves could be operated by the central bank under the **ECB Pontes model**, or by a trusted third party under the **Swiss Helvetica model**, but would only benefit use cases where tokenized assets are natively maintained on this same trusted platform and thereby able to settle atomically with legal finality. Where the tokenized asset settles against any other money token or synchronizer, the tokenized reserve adds no special benefit but introduces the complication of multiple liquidity pools.

In the short term, a fully-featured RITS synchronization operator facility with appropriate finality designation should be able to deliver programmability, operating hours and liquidity efficiency for most practical use cases.

Q13 · Functional outcomes

A tokenised reserve design should support **atomic settlement**, smart contracts, collateralised funding, continuous liquidity management, and external connections that allow heterogenous asset tokens to be minted on the ledger with legal effect. This multi-asset programmable ledger would be maintained either by the RBA (on the Eurosystem model) or by a trusted national industry user-owned operator (such as AP+, on the Swiss model). National scheme rules would define the authoritative delivery record and govern cross-platform transaction recovery.

Area 4: Tokenised reserves *(continued)*

Q14 · Issuance and distribution

The ideal model would involve a "**jurisdictional ledger**" maintained on an open-source platform (e.g. **Hyperledger Besu**), owned and operated either by the RBA or by a national industry consortium (i.e. AP+) under RBA oversight, able to support same-ledger **atomic settlement** between wholesale tokenized reserves and heterogenous tokenised assets – potentially including natively minted Australian Government Securities. This hypothetical ledger – which could effectively function as a **multi-asset RITS-Austraclear-DLT** – would be consistent with the **Agora framework** for maintaining an international system of two-tiered money.

Q15 · Funding, redemption, and liquidity

Conversion between ESA balances and tokenised reserves is unnecessary where the tokenised reserves are "**digital twin**s" (i.e. ESAs are locked against circulating tokens). The RBA should directly issue tokens against these locked reserves rather than encourage RBDC issuance. The RBA would issue tokenised reserves against locked reserves (as per Agora), avoiding the liquidity fragmentation that could result from "real" wCBDC being sold for reserves in the manner of physical currency. Funding, redemption, and liquidity-management concerns arise where operating hours of systems differ, recommended extended operating hours for the main RITS settlement engine. Funding must be accessible across systems to avoid fragmenting liquidity in separate reserve pools.

Australian Payments Plus Impact Assessment

AP+ in a tokenised ecosystem

Tokenisation offers each AP+ service capabilities that their existing rails cannot supply

	TODAY	WITH TOKENISATION
BPAY	Bill payments clear through BPAY, but banks' net obligations still settle in a daily BECS batch, the one BECS use case the NPP cannot yet carry; volumes flat. **Osko**'s expansion has stalled, and merging it with SCT is a roadmap item.	Biller codes and reconciliation travel with the token; each bill settles on its reference, no BECS. **Osko**'s immediate credit and payment status live in the transfer itself. A token-borne biller code could also be presented from outside Australia, opening BPAY to international payers.
eftpos	Owners' incentives are mixed; the scheme competes on fee cuts and routing rules; card-not-present is where it is weakest.	Token settlement behind existing acceptance, as Visa and Mastercard do with stablecoins: eftpos keeps authorisation, fraud and dispute rules and gains support for cross-border payments.
NPP	Wholesale fees well above BECS; the 2030 BECS end date dropped; new connections lag commitments. **PayTo** has yet to mature as a direct debit replacement.	Retirement of the Swift mesh network (and its high running costs) with an upgraded FSS. Escrow, split payments and delivery against payment. A **PayTo** mandate becomes a standing, revocable spend authority on the ledger.
ConnectID	Consented identity exchange between banks and businesses; no payment authority; available to customers of the four majors.	The gate to any token ecosystem: verified identity for wallets, issuers and platforms, with personal data kept off the ledger and payment authority still separate.

Sources: RBA, BECS decommissioning risk assessment, March 2026 (NPP fees, PayTo maturity); AusPayNet, 16 Dec 2025 (end date dropped); AP+, BECS migration (BPAY settlement); AP+ roadmap (Osko); ACCC, 2021 (eftpos incentives); AP+ ConnectID.

New Payments Platform

	A stranded asset	No reservations or batches	Priced per item
Today	A Swift-built mesh of bilateral gateways that tokenisation makes obsolete. Swift remains its vendor, and the RBA expects its wholesale costs to stay above BECS (RBA, 2025).	The FSS settles or rejects, with no reservation, queue or liquidity-saving mechanism. The NPP has no bulk message, so there are no batches.	Two cents each side, well above BECS. The per-item tariff is how the build and the Swift network are being paid off, so bulk files of small payments pay the most.
On a Ledger	Each participant runs a node over ordinary network connections. There is no vendor network and no gateway to pay for.	A transfer is locked before it settles, so reservation is part of the ledger. A batch is a set of transfers that commit together or not at all.	No vendor network charging by the message. Operator build recoverable through membership rather than per-item tariff.

RAIL	TOPOLOGY	2025 VOLUME	PUBLISHED CHARGE
NPP	Mesh	1.86bn	**A$0.04 combined**
TIPS	Hub	2.47bn	**A$0.003 combined**
Pix / SPI	Hub	79.8bn	**A$0.0003 per credit**
FedNow	Hub	8.4m	**A$0.065, sender pays**
Faster Payments	Hub	5.5bn	**A$0.029 send + receive**
SIC	Hub	1.03bn	**A$0.035 average**

Members are building on-ramps…

Cuscal issued a stablecoin in Project Acacia with the NPP as on- and off-ramp; Westpac settled tokenised assets via PayTo. Both are now developing NPP ramps to programmable token networks.

…but who will want the off-ramp?

NPP can be the on-ramp to a **global network of programmable ledgers**: instant payments across borders with negligible fees. Once there, funds can keep circulating through payments, foreign exchange and asset transactions. If recipients accept and reuse those tokens, users need an off-ramp only when they need a bank-account balance.

Sources: RBA, BECS decommissioning risk assessment, March 2025, p. 17 (NPP wholesale costs); AP+, Move to NPP (Swift as vendor); RBA, RITS consultation, ch. 2 (FSS); Cuscal, 16 Feb 2026; Project Acacia report (Westpac, PayTo); ACCC determination summary, 2021. Volumes: RBA C6.1 (NPP, 2025); ECB TARGET annual report 2025; Banco Central do Brasil; FRB FedNow statistics; Pay.UK annual statistics 2025; SIX Interbank Clearing, financial statements 2025 (SIC: 2.02 centimes average).

eftpos

Existing rails

The fee gap
eftpos is the cheaper debit network for most merchants, which is why the RBA expects least-cost routing.

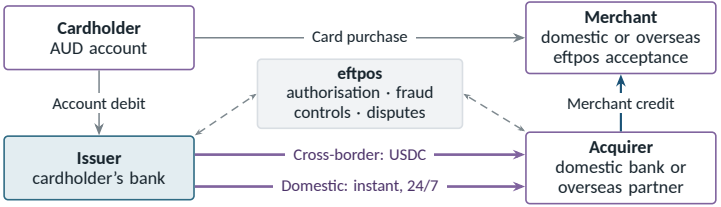
Domestic only
Overseas purchases on dual-network cards use Visa or Mastercard; eftpos is domestic.

Daily bank settlement
Interbank eftpos obligations settle in one net RITS batch per business day.

An eftpos stablecoin?
eftpos can use existing USD stablecoin arrangements, like Visa and Mastercard's, to offer cross-border payments. Domestically, an eftpos token on a new domestic ledger could support 24/7 settlement and programmability.

Going global with stablecoin settlement

The first least-cost route for Australian cards internationally: eftpos fees instead of international scheme fees, with eftpos authorisation, fraud controls and dispute rules retained.



Cross-border: the token is converted to USDC and received by the overseas acquirer, which pays the merchant in USDC or local currency.
Domestic: money token transfers settle 24/7 instead of daily weekday batching.

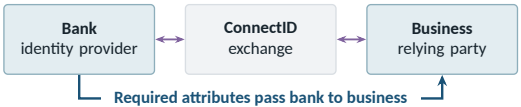
Conversion is done by the issuing bank, or by eftpos through a licensed stablecoin issuer. Whoever converts holds the FX and reserve risk. A token transfer is final, so a chargeback is a new reversing transfer funded by the acquirer under existing scheme rules.

Visa and Mastercard support stablecoin settlement between issuers, schemes and acquirers beyond banking hours, with existing card protections. Visa's Australian Crypto.com programme uses USDC cross-border.

ConnectID

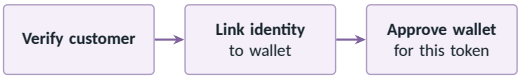
A bank-verified identity that can be bound to a wallet without putting personal data on the ledger.

How it works today



The customer chooses their bank, logs in there and consents; only the attributes the business asked for are released. ConnectID routes the request and never sees or stores the data. Built on **OpenID Connect** and **FAPI 2.0**, with the four majors as identity providers.

Proposed token application



Approval can expire or be withdrawn

The ledger records only that this wallet belongs to a person or firm a named bank has verified, with an expiry and a revocation path. Name, date of birth and documents stay with the bank. **Token-level permissions of this kind are how Agorá's Noto contracts enforce issuer rules.**

USES IN TOKENISED PAYMENTS

Verify wallet customers. Check a customer's identity through their bank when they join a token service.

Control token access. Link that identity check to a wallet the issuer permits to hold or receive its tokens.

Support transfer checks. Supply verified customer details where information about the sender and recipient is required.

OTHER CHECKS REQUIRED

Wallet control, signing keys, corporate mandates, sanctions screening and due diligence remain separate checks, and personal data stays off immutable ledgers.

EXISTING USES

Confirming someone's age (Persona, April 2026). Verifying the developer registering AI agents (AstraSync, June 2026).

STATUS

Accredited since September 2021 under the **Trusted Digital Identity Framework (TDIF)**, the first non-government exchange. Now accredited under the Digital ID Act 2024. Live since October 2023; identity providers are CommBank, NAB, ANZ Plus and Westpac. Joining the government system requires separate approval; private-sector participation opens by December 2026.

Sources: AP+ ConnectID and launch, 16 Oct 2023; OpenID Foundation, FAPI 2.0 certification; Biometric Update, 1 Apr 2026 (ACCC register); Allens, Apr 2026 (AGDIS phases); AUSTRAC transitional rules (travel rule); Agorá report, p. 36 (Noto).

A new national ledger

The national ledger would be Australia's **Agorá jurisdictional ledger**, built and operated by **AP+** on **Hyperledger Besu** under RBA rules. Australian banks mint **AUD deposit tokens**, eligible tokenised assets are recorded, and **Australian domestic payments clear on domestically controlled infrastructure**. The RBA alone issues and redeems tokenised ESA balances (mirrored ESA funds used for deposit token interchange) **on this same ledger**; its settlement trigger is final under RITS rules and legal finality designation.

NPP today

Osko and PayTo clear over a mesh of bilateral gateways that Swift built and operates, then settle in the FSS. Vendor costs of tens of millions per year.

eftpos today

Card authorisation and clearing through the scheme; issuer and acquirer settle net in RITS by batch.

BPAY today

Bill payments clear through BPAY; net settlement uses BECS.

reroute

AP+ Ledger: Australia's Agorá jurisdictional ledger

Bank integration software would translate existing NPP, eftpos and BPAY instructions into token-payment calls on the AP+ ledger.

Banks mint and redeem **AUD deposit tokens**; issuers and depositories keep legal registers for **eligible tokenised assets**.

An AP+ payment coordinator contract would collect bank confirmations for each payment and authorise linked deposit and reserve transfers once funds are locked.

confirm payee → validate → lock → settle

Tokenised ESA balances. Interbank obligations from deposit-token payments settle here gross, leg by leg; eftpos-style batches could settle net.

The RBA alone issues, redeems and sets eligibility; RITS and the FSS stay the account system and point of finality.

No AUD obligation needs to route over an offshore orchestration layer or settle in a venue the RBA does not oversee.

Why Besu? Open-source, EVM-compatible software used in Agorá's prototype and Swift's shared-ledger MVP. It supports established Ethereum smart-contract tooling and token standards, and permissioned QBFT consensus.

APPENDIX

Regimes

Canada

1 Messaging / connectivity
- Swift → Lynx, CLS; ● CSN → ACSS / USBE
- Interac network → e-Transfer, IMN debit

2 Wholesale payments / FX
- Lynx (● Payments Canada) OP ● IBM TECH ● Nexi NET ● Swift
- CLS: international bank-owned group

3 Retail payment systems
- ACSS (● Payments Canada; TECH ● IBM); ● Interac
- Visa / ● Mastercard (US listed groups)

4 CCPs by instrument

Cash equities: ● CDS; derivatives / repo: ● CDCC
Both under ● TMX; CDS TECH ● TCS BaNCS (2025)
OTC rates: ● LCH / ● LSEG (UK)

5a Sovereign-debt CSD
- CDS / ● CDSX (● TMX) TECH ● TCS; cash via ● Lynx

Government of Canada securities

5b Other CSD / securities settlement
- CDS / ● CDSX (same CSD)

Other debt, equities and funds

6 Central-bank settlement
- Bank of Canada: public reserve authority
- Lynx processing: ● Payments Canada (OP ● IBM)

Lynx: CAD 371.9bn and 54,229 payments per business day, 2025.

Owner / operator / technology / settlement

Lynx: Payments Canada (statutory member non-profit) owns the scheme and is formal operator; IBM Canada is contracted for hosting, integration and daily operations (live service); the application was built and is supported by Nexi (ex-SIA); messaging over Swift; settlement in Bank of Canada accounts. Hosting location and ability to run without IBM / Nexi: unknown. ACSS / USBE run over domestic CSN; IBM was technology partner for the 2020 refresh. Interac is user-owned; suppliers undisclosed. TMX is listed; CDS moved to TCS BaNCS in April 2025 (installed, vendor-supported).

CDSX settlement

CDSX links securities and cash-account transfers. Net CAD obligations discharge through Lynx into CDS's BoC account. Individual trades do not each settle gross in reserves.

Project Agorá

Bank of Canada joined on 27 May 2026, after the initial prototype had been completed.

● Domestic public
● External industry / user-owned
● Domestic industry / user-owned
● External corporate
● Domestic commercial

United States



1 Messaging / connectivity
- FedLine → Fedwire / FedNow / FedACH
- SMART (● DTCC); ● Swift → CHIPS, CLS (cooperative)

2 Wholesale payments / FX
- Fedwire Funds; ● CHIPS (● TCH; settles at ● Fed)
- CLS: US operator, international bank owners

3 Retail payment systems
- FedNow / ● FedACH; ● RTP (● TCH; TECH ● Mastercard)
- EPN (● TCH); ● Visa / ● Mastercard (US listed groups)

4 CCPs by instrument

Treasury / repo / MBS: ● FICC; Treasury also ● ICE
Equities: ● NSCC; derivatives: ● CME; options: ● OCC
Credit: ● ICE; OTC rates also clear at ● LCH (UK)

5a Sovereign-debt CSD
- Fedwire Securities (● Federal Reserve)

Treasuries and eligible agency securities

5b Other CSD / securities settlement
- DTC (● DTCC); funds via settling banks at ● NSS

Equities, corporate debt and other securities

6 Central-bank settlement
- Federal Reserve accounts
- Fedwire Funds, ● FedNow and ● NSS

Fedwire: USD 4.593tn / 869,187 payments daily; CHIPS: USD 2.014tn / 632,419, 2025.

Owner / operator / technology / settlement

Fedwire, FedNow, FedACH, Fedwire Securities: Reserve Bank-owned, -run and -built. RTP: TCH on Mastercard / Vocalink software; prefunded joint Fed account. DTC / NSCC / FICC: DTCC in-house over its SMART network; funds via NSS. CME / ICE in-house. CLS: Swiss bank-owned; IBM hosts CLSSettlement; Swift messaging. TCH / DTCC unlisted, CME / ICE listed, OCC exchange-owned; Fed member stock carries statutory rights. Q2 2026: RTP US$576bn / 142m payments, FedNow US$274.7bn / 5m.

DTC / BNY

July 2026 production trades used digital twins of DTC-held securities. Twins run on Besu and Canton (Digital Asset); DTC remains the register. BNY keeps the official fund records; Goldman Sachs DAP holds mirror tokens.

Bank payment platforms

Kinexys (listed JPMorganChase): over US$4tn cumulative and US$7bn average daily across the platform, August 2026. Customers' cubiX: US$1.5tn instant transfers, Oct 2024–Q2 2026, with US$5tn across all rails.

Other asset / cash platforms

Figure records HELOC ownership and liens on Provenance. Q2 2026 marketplace volume was US$4.3bn. WA VenueX (Western Alliance) connects bank-dollar settlement and stablecoin services.

● Domestic public
● External industry / user-owned
● Domestic industry / user-owned
● External corporate
● Domestic commercial

United Kingdom



1 Messaging / connectivity

● Swift → CHAPS, CREST; BT ● Settlenet → CREST
● Vocalink (US) networks → Bacs / ICS / FPS

2 Wholesale payments / FX

● CHAPS (● BoE RT2; built by ● Accenture, in-house)
● £FnPS (● Fnality); ● CLS (TECH ● IBM)

3 Retail payment systems

● FPS / ● Bacs / ● ICS (● Pay.UK) OP+TECH ● Vocalink
● LINK (OP ● Vocalink); ● Visa / ● Mastercard (US)

4 CCPs by instrument

Rates / repo / equities: ● LCH (● LSEG 94.4%)
Derivatives: ● ICE (US IT); metals: ● LME Clear
● LME parent ● HKEX; TECH Cinnober (● Nasdaq)

5a Sovereign-debt CSD

● CREST / ● Euroclear UK & International: gilts
● Euroclear: Belgian parent; DvP in ● BoE money

5b Other CSD / securities settlement

● CREST (same CSD)
Equities and other eligible securities

6 Central-bank settlement

● Bank of England RTGS (RT2, live April 2025)
Public sterling reserve authority

CHAPS: GBP 371.3bn and 210,482 payments per business day, 2025.

Owner / operator / technology / settlement

CHAPS / RTGS: BoE owns and runs RT2; Accenture built it (handover completed 2026; BoE-hosted); external dependencies include the Swift network. FPS / Bacs / ICS: Pay.UK owns the schemes; Vocalink (Mastercard, US) designs, builds and operates them under contracts to the early 2030s; hosting unknown; net settlement in RTGS. CREST: Euroclear UK & International (Belgian unlisted parent) runs it in the UK via Swift CRNet or BT Settlenet; DvP in central bank money. ICE Clear Europe takes IT from US ICE entities; LME Clear runs Cinnober (Nasdaq); CLS is hosted by IBM. Pay.UK and LINK are non-profits; BoE controls sterling reserves.

DIGIT: digital sovereign debt

HM Treasury's **Digital Gilt Instrument** pilot will issue short-dated government debt on HSBC Orion within the Digital Securities Sandbox, with the first issuance expected by **Q1 2027**. Joint lead managers were appointed on 6 October 2026. A planned link connects Orion as issuer Digital Securities Depository (DSD) to LSEG B3 as investor DSD.

The stated objective includes developing UK-based DLT infrastructure. CREST is a UK CSD under a Belgian parent. *A possible strategic benefit is a UK-headquartered alternative to Euroclear for sovereign-debt issuance and settlement* (replacing CREST is not an announced objective).

● Domestic public
● External industry / user-owned
● Domestic industry / user-owned
● External corporate
● Domestic commercial

Eurosystem

1 Messaging / connectivity
- ESMIG → T2 / T2S / TIPS via ● Swift or ● Nexi (IT)
- ● SwiftNet / ● SIAnet / EBICS → STEP2, RT1; ● RNI

2 Wholesale payments / FX
- T2 (● Eurosystem; OP+TECH ● 4CB); ● CLS
- ● EURO1 (● EBA CLEARING) OP+TECH ● Swift

3 Retail payment systems
- ● TIPS; ● STEP2 / ● RT1 (● EBA CLEARING) OP ● Nexi
- ● STET / ● CORE (in-house); ● Visa / ● Mastercard (US)

4 CCPs by instrument
- ● Eurex: derivatives / repo (DE parent)
- ● Euronext: cash / derivatives (NL parent)
- ● LCH (UK), ● Cboe (US), ● BME (Swiss ● SIX) parents

5a Sovereign-debt CSD

National issuer CSDs; Belgium: ● NBB-SSS
Ireland: ● Euroclear Bank; Spain: ● Iberclear / ● SIX

5b Other CSD / securities settlement
- ● Euroclear, ● Clearstream, ● Euronext and other CSDs
- ● T2S: shared settlement engine (OP+TECH ● 4CB)

6 Central-bank settlement
- ● T2, ● TIPS and ● T2S cash services
- ● Eurosystem central-bank accounts

T2: EUR 1.933tn and 431,067 payments per business day, 2025.

Owner / operator / technology / settlement

Euro-area scope. T2 / T2S / TIPS: Eurosystem-owned, developed and operated by the 4CB (Banca d'Italia, Banco de España, Banque de France, Bundesbank); access only via ESMIG providers Swift or Nexi under ten-year concessions. EURO1: EBA CLEARING (48 bank shareholders) owns the scheme; Swift develops, maintains and processes it (software and live service). STEP2 / RT1: technical provider and operator Nexi Payments (Italy; listed parent Nexi S.p.A.); SwiftNet, SIAnet or EBICS access. STET runs its own CORE platform; Eurex clears on Deutsche Börse's in-house C7. Deutsche Börse / Euronext are listed; Euroclear unlisted and institutionally owned. LCH SA has a UK parent. Swiss SIX owns Iberclear / BME. National CSDs retain their functions while T2S supplies shared securities settlement.

Pontes: live interoperability

Launched 21 September 2026, linking market DLT platforms to TARGET. Axiology, Cashlink, Clearstream and SWIAT were onboarded. Initial Hash-Link settlement uses T2 cash. Securities remain on market platforms.

Appia: integrated tokenised markets

Blueprint due in **2028** for tokenised wholesale markets combining central-bank money, private money, and securities. Appia explores shared or interoperable DLT networks, common standards, and European governance to reduce fragmentation and foreign dependencies. Its findings will guide Pontes enhancements.

● Domestic public
● External industry / user-owned
● Domestic industry / user-owned
● External corporate
● Domestic commercial

Switzerland

1 Messaging / connectivity
- SSFN (● SNB / ● SIX; SCION ● Anapaya) → SIC
- Swift → SIC RTGS; SIC instant since Nov 2025

2 Wholesale payments / FX
- SIC: ● SNB mandate; OP+TECH ● SIX (SIC Ltd, 100%)
- CLS: international bank-owned group

3 Retail payment systems
- SIC instant payments; ● TWINT (OP ● SIX)
- Visa / ● Mastercard (US listed groups)

4 CCPs by instrument

Cash securities: ● SIX x-clear
Derivatives: ● Eurex (DE), ● LCH (UK)
- SIX / ● BME integration remains an announced plan

5a Sovereign-debt CSD
- SIX SIS / ● SECOM: Swiss government securities
- SIX: financial-institution-owned Swiss group

5b Other CSD / securities settlement
- SIX SIS / ● SECOM; ● DAP (ex-SDX; TECH ● R3 Corda)

Other securities and international custody links

6 Central-bank settlement
- SNB accounts; ● SIC run by ● SIX Interbank Clearing
- SNB: statutory central bank with listed shares

SIC: CHF 220bn / 4.1m payments per business day, 2025; includes retail.

Owner / operator / technology / settlement

SIC: SNB is system manager; SIX Interbank Clearing (100% SIX; SNB board seat; PostFinance holds SIX shares, not SIC) builds the software and runs two Swiss data centres. **Unusually, the national RTGS has a private technical operator.** The SNB determines access, supplies liquidity, and sets settlement rules; payments settle in SNB money. Access via ● SSFN (SCION from Swiss Anapaya; carriers include BT Radianz) or Swift. SIX unlisted, user-owned (foreign banks 14%). SIX SIS vendor undisclosed; DAP (ex-SDX) built on R3 Corda. TWINT: SIX processing; Worldline minority. Derivatives and cards add foreign dependencies.

SeturionX: production SIC link

BX Digital is now SeturionX, a Swiss-regulated firm in Germany's privately held Börse Stuttgart group. Public Ethereum ledger; payment module built with GFT. SNB confirms production use of the SIC link. The link synchronises an external asset ledger with conventional CHF settlement, separately from Helvetia's cash token.

SIX / UBS

SIX combines digital and conventional securities in SIX SIS. UBS Tokenize includes live securities and uMINT. Platform ownership and the issuer of the settlement claim remain separate.

● Domestic public
● Domestic industry / user-owned
● Domestic commercial
● External industry / user-owned
● External corporate

Singapore



1 Messaging / connectivity
- Swift → MEPS+ (sole access channel), CLS
- BCS (● NETS) network → FAST / PayNow / GIRO

2 Wholesale payments / FX
- MEPS+ (● MAS owns, operates; vendor undisclosed)
- CLS: international bank-owned group

3 Retail payment systems
- FAST / ● PayNow / ● GIRO (● SCHA / ● ABS → ● SPaN)
- OP ● BCS TECH ● Vocalink; ● NETS; ● Visa / ● Mastercard

4 CCPs by instrument

Cash securities: ● CDP (TECH ● LSEG Millennium)
Derivatives: ● SGX-DC (TECH ● Nasdaq Genium)
Both under ● Singapore Exchange (listed)

5a Sovereign-debt CSD
- MEPS+ securities (● MAS)

Primary Singapore government-securities function

5b Other CSD / securities settlement
- CDP (● SGX) TECH ● LSEG Millennium PostTrade

Other securities; customer government holdings

6 Central-bank settlement
- MAS accounts through ● MEPS+

Public SGD reserve authority

Owner / operator / technology / settlement

MEPS+: MAS owns, operates and hosts; an unnamed external vendor supplies and maintains the software; Swift is the sole disclosed access channel. FAST / PayNow / GIRO: schemes of SCHA and ABS, moving to ● SPaN (MAS / ABS guarantee company; members MAS and seven D-SIBs, four foreign-owned) by end-2026; BCS (NETS: DBS / OCBC / UOB equally) operates them in its own data centres on Vocalink (US Mastercard) software; live vendor-support need unknown; net settlement in MEPS+. PayNow is FAST addressing. CDP runs LSEG Millennium PostTrade, SGX-DC Nasdaq Genium; SGX listed. MAS runs reserves / government-securities settlement.

Partior / DBS / Whale

Partior is a privately held Singapore company with international bank / institutional investors; no controlling shareholder identified (control unresolved). At least five banks were live by July 2026, including EUR and USD routes. These transfer commercial-bank money. DBS Token Services is commercial. Ant's Whale treasury services overlap bank-operated routes.

● Domestic public
● External industry / user-owned
● Domestic industry / user-owned
● External corporate
● Domestic commercial

Hong Kong

1 Messaging / connectivity
- Swift → HKD / USD / EUR / RMB CHATS, CMU (part)
- HKICL network → FPS; SDNet/2 (HKEX) → CCASS

2 Wholesale payments / FX
- CHATS HKD / USD / EUR / RMB (OP HKICL)

SETTLE HKMA; HSBC; StanChart; BOCHK; CLS FX

3 Retail payment systems
- FPS (HKICL; own network); Octopus (MTR)
- Visa / Mastercard (US listed groups)

4 CCPs by instrument

Equities: HKSCC (in-house CCASS); options: SEOCH
Futures: HKCC (TECH Nasdaq; in-house ODP by 2028)
OTC rates: OTC Clear (92.5% HKEX; TECH Calypso)

5a Sovereign-debt CSD
- CMU OmniClear: 80% Exchange Fund / 20% HKEX

OP HKICL computer operator; Swift channel

5b Other CSD / securities settlement
- CMU: other debt; CCASS / HKSCC: equities
- HKEX: government stake / board rights

6 Central-bank settlement

HKD: HKMA money via HKD CHATS / FPS
USD / EUR / RMB: commercial-bank settlement

Owner / operator / technology / settlement

CHATS: HKMA owns HKD CHATS and is its settlement institution; HKICL (50% HKMA / 50% Hong Kong Association of Banks) operates all four CHATS, FPS and the CMU systems and handles development and maintenance (vendor undisclosed). Messaging on Swift since 2009. USD, EUR and RMB settle in commercial-bank money at HKMA-appointed HSBC, Standard Chartered and Bank of China (Hong Kong) (UK / mainland parents). FPS runs on HKICL's own network. CMU: HKMA remains owner and system operator; CMU OmniClear (80% Exchange Fund / 20% HKEX) runs operations. HKEX derivatives clear on Nasdaq Genium until the in-house ODP (2027–28). Octopus is 64.02% owned by listed, state-majority MTR.

Orion / mBridge

The 2024 government digital bonds used HSBC Orion deployed to CMU, with definitive records maintained by CMU. HKMC issued HKD12bn digitally on Orion in June 2026. Cash arrangements vary by issue. mBridge is a separate multi-CBDC cross-border platform, co-governed with the PBoC Digital Currency Institute, Bank of Thailand and CBUAE since the BIS handover.

Domestic public
Domestic industry / user-owned
Domestic commercial
External industry / user-owned
External corporate

China

1 Messaging / connectivity
- CNFN (● CNCC) → HVPS / BEPS / IBPS; ● CIPS
- Swift via ● FGIS JV (Swift 55%; ● CNCC, ● CIPS)

2 Wholesale payments / FX
- HVPS (● PBOC; OP+TECH ● CNCC); ● CIPS cross-border CNY outside ● CLSSettlement

3 Retail payment systems
- BEPS / ● IBPS; ● UnionPay; ● NetsUnion (● NUCC)
- Mastercard–NUCC JV (51% ● Mastercard); ● AmEx JV

4 CCPs by instrument
- ChinaClear: exchange securities (state-controlled)
- SHCH: OTC rates / FX / credit (state-owned)
- CFFEX: financial futures (state-controlled exchanges)

5a Sovereign-debt CSD
- CCDC (state-owned)

Core interbank government-bond CSD

5b Other CSD / securities settlement
- ChinaClear: exchange securities
- CCDC / ● SHCH: eligible interbank debt

6 Central-bank settlement
- HVPS / ● PBOC accounts

Public domestic reserve authority

HVPS: CNY 35.57tn and 1,596,800 transactions per business day, Q2 2025.

Owner / operator / technology / settlement

HVPS / BEPS / IBPS: PBOC-owned, operated by its public institution CNCC on the domestically built CNAPS2; no Swift. CIPS supplies messaging, clearing and settlement. Swift connectivity runs through FGIS, a Beijing JV in which Swift holds 55%. NetsUnion (NUCC) clears online payments; PBOC-system entities hold 37%. Alipay / WeChat serve customers. ● Mastercard NetsUnion is a separate bank-card clearing JV between Mastercard (51%) and NUCC, operating since May 2024.

e-CNY: bank liabilities

From January 2026, commercial-bank wallets are deposit liabilities with interest and deposit insurance. Reserve requirements apply to bank wallets. Non-bank issuers maintain 100% backing. e-CNY combines accounts and smart contracts.

International Operation Centre

PBOC's Shanghai centre opened in September 2025, covering cross-border payments, blockchain services and digital assets. Chengfang Chain supports on-chain settlement, cross-chain links, issuance, registration and custody.

mBridge

Central banks issue money on a shared DLT for cross-border transfers and FX PvP. Tokenised cash settles on-platform, separately from HVPS; co-governed by five central banks since the BIS handover (October 2024). Bank of China: over RMB600bn cumulative, April 2024–June 2026.

● Domestic public
● External industry / user-owned
● Domestic industry / user-owned
● External corporate
● Domestic commercial

Japan

1 Messaging / connectivity
- BOJ-NET network; Zengin-Net (TECH NTT Data)
- JASDEC network; arrownet (JPX); Swift intl

2 Wholesale payments / FX
- BOJ-NET Funds; FXYCS (OP BOJ, in BOJ-NET)
- CLS: international bank-owned group

3 Retail payment systems
- Zengin (OP+TECH NTT Data); JCB (unlisted)
- Visa / Mastercard (US listed groups)

4 CCPs by instrument
- JSCC (JPX): cash, JGB / repo, derivatives
- JDCC (JASDEC): non-exchange DvP; TFX: futures
- TFX industry-owned; LCH; JSCC TECH Nasdaq

5a Sovereign-debt CSD
- BOJ-NET JGB Services
- Bank of Japan sovereign-securities function

5b Other CSD / securities settlement
- JASDEC: other debt, equities and funds

Unlisted company with industry shareholders

6 Central-bank settlement
- BOJ-NET Funds; statutory central-bank control

55% state capital; listed subscription certificates

BOJ-NET Funds: JPY 232.6tn and 94,602 transactions per business day, 2025.

Owner / operator / technology / settlement

BOJ-NET: BOJ designs, hosts and operates; contractors built it under BOJ control; participant gateways from NTT Data and others. Zengin: Zengin-Net (Japanese Bankers Association) owns the scheme; NTT Data develops and maintains it (installed software plus support; the October 2023 outage was an NTT Data defect). JSCC outsources IT to TSE; derivatives engines (Cinnober, Calypso) are Nasdaq-owned. BOJ certificates carry no management rights. JASDEC industry-owned, JPX listed, JCB privately held. Zengin payments of at least JPY100m settle gross; smaller net.

Progmat / ODX / BOOSTRY

Progmat migrated all its security tokens, over JPY452bn, to an Avalanche L1 in July 2026. Platform software moved from R3 Corda to Ava Labs; validator operators unknown; trust banks keep the legal register. ODX's START secondary market opened in December 2023. BOOSTRY reports cumulative issuance of JPY333.3bn to March 2026.

Stablecoins / deposit tokens

The June 2023 regime permits eligible banks, trust companies and funds-transfer providers to issue. JPYC launched in October 2025. DCJPY bank deposit tokens settle environmental-value trades.

- Domestic public
- Domestic industry / user-owned
- Domestic commercial
- External industry / user-owned
- External corporate

South Korea

1 Messaging / connectivity
- BOK-Wire+ closed network; KFTC network
- Stock-Net (Koscom / KRX); Swift international

2 Wholesale payments / FX
- BOK-Wire+ (Bank of Korea; built by LG CNS)
- CLS for KRW FX: international bank-owned

3 Retail payment systems
- KFTC electronic banking / giro
- BC Card (KT parent), Visa / Mastercard

4 CCPs by instrument
- KRX: cash securities and listed derivatives

Eligible OTC rates; TECH Koscom (76.6% KRX)
Unlisted, financial-industry shareholders

5a Sovereign-debt CSD
- Korea Securities Depository (KSD; 70.4% KRX)

Next system by LG CNS; DvP in BOK-Wire+

5b Other CSD / securities settlement
- KSD (same CSD)

Other bonds, equities and funds

6 Central-bank settlement
- BOK-Wire+ (Bank of Korea)

Statutory public reserve authority

Owner / operator / technology / settlement

BOK-Wire+: owned and run by BOK; third-generation system built 2018–20 by an LG CNS consortium (support terms unknown); no Swift domestically. KFTC: member non-profit (ten general-member banks including BOK) on its own shared network; vendors undisclosed. KRX / KSD: unlisted industry companies; clearing systems from group-owned Koscom; KSD's next-generation system by LG CNS (phase 1 live July 2026). No foreign core supplier found for any Korean core FMI; CLS and card schemes are the external services. Fast KFTC customer transfers generate interbank net obligations in BOK-Wire+, rather than continuous gross reserve settlement.

Project Hangang: a national deposit-token network

The BOK leads the national multi-bank pilot and operates its shared ledger. Retail customers use commercial-bank deposit tokens; banks settle between themselves in **BOK wCBDC on the same platform**. Cross-bank payments burn the payer bank's tokens and issue the recipient bank's tokens at par, coordinated with central-bank-money settlement.

The April–June 2025 live pilot involved seven banks, about 80,000 users and 12,000 merchants. Phase 2 expands to nine banks and programmable government disbursements: a practical implementation of the BIS unified-ledger model.

Governor **Hyun Song Shin**, appointed in April 2026, previously served as BIS Economic Adviser and Head of the Monetary and Economic Department. An advocate of the unified ledger at the BIS, he now leads the central bank implementing it.

Domestic public
Domestic industry / user-owned
Domestic commercial
External industry / user-owned
External corporate

Brazil

1 Messaging / connectivity
RSFN: ● RTM (● ANBIMA / ● B3) and ● Embratel carriers → STR, SPI, Selic, Nuclea, B3; ● Swift intl links

2 Wholesale payments / FX
● STR (● BCB in-house); ● SITRAF (● Nuclea)
● B3 FX clearing; BRL outside ● CLSSettlement

3 Retail payment systems
● Pix / ● SPI (● BCB builds, runs); ● SILOC (● Nuclea)
● Elo (TECH ● Microsoft cloud); ● Visa / ● Mastercard

4 CCPs by instrument
● B3 Clearinghouse: securities and derivatives
● B3 Foreign Exchange Clearinghouse: FX
● B3 listed; TECH Cinnober (● Nasdaq) licence

5a Sovereign-debt CSD
● Selic: federal government securities
● BCB operates; ● ANBIMA supplies mainframes

5b Other CSD / securities settlement
● B3 depository / OTC infrastructure (former ● Cetip)
Equities and other debt; listed ● B3 parent

6 Central-bank settlement
● STR and ● SPI
● Banco Central do Brasil accounts

Owner / operator / technology / settlement

STR and SPI / Pix: BCB owns, operates and supplies the IT; SPI access through the RSFN, two homologated carrier networks run by RTM (ANBIMA 80% / B3 20%) and Embratel (Claro); STR also offers STR-Web internet access. Selic: BCB with ANBIMA, which finances the mainframes. Nuclea is an unlisted financial-institution-owned company; vendors undisclosed; settles via STR. B3 is listed; its clearing runs Cinnober (Nasdaq) under a perpetual licence, and its FX clearing still depends on foreign-currency settlement banks. Elo is a commercial company owned by three Brazilian banks (two federally controlled), moving its processing to Microsoft cloud. BRL has no CLSSettlement route.

Pix / SPI and securities DvP

Pix is the payment scheme. SPI settles interbank Pix payments individually, 24/7, in BCB money. Some transfers settle within one participant's books. Selic and B3 link securities delivery to cash settlement through their respective arrangements.

● Domestic public
● External industry / user-owned
● Domestic industry / user-owned
● External corporate
● Domestic commercial

Australia



1 Messaging / connectivity
- Swift PDS → HVCS; Swift NPP network → NPP
- COIN (AusPayNet; OP TNS) → BECS, cards, BPAY

2 Wholesale payments / FX
- HVCS (AusPayNet) NET Swift PDS SETTLE RITS
- CLS for AUD FX: international bank-owned group

3 Retail payment systems
- NPP (AP+) OP+TECH Swift SETTLE FSS
- eftpos / BPAY (AP+); BECS; Visa / Mastercard

4 CCPs by instrument
- ASX Clear: equities (TECH TCS, AWS) / options
- ASX Clear (Futures): futures (TECH Nasdaq)

Offshore rates: LCH (UK), CME (US)

5a Sovereign-debt CSD
- Austraclear (ASX; EXIGO, supported in-house)

Commonwealth and other government debt

5b Other CSD / securities settlement
- Austraclear: other debt; CHESS: equities
- Clearstream (LU) / Euroclear Bank (BE): extra access

6 Central-bank settlement
- RITS System Queue / FSS (RBA)

Statutory public reserve authority

RITS ex FSS: AUD 288bn and 69,000 payments per business day, FY2024/25.

Owner / operator / technology / settlement

NPP: NPPA (AP+) owns the scheme and Basic Infrastructure; Swift designed, built and operates it under a long-term contract on a dedicated domestic Swift network; settles in the RBA's FSS. Hosting and exit terms: unknown. COIN: AusPayNet administers, TNS (US, private) operates. HVCS: Swift PDS messaging, RITS settlement. RITS / FSS: RBA-run, no vendor named. ASX (listed): equity clearing on TCS BaNCS / AWS since April 2026; futures on Nasdaq Genium; Austraclear's EXIGO supported in-house; CHESS settlement replacement 2029. AP+ unlisted, industry-owned; AusPayNet non-profit. Clearstream licensed; Euroclear Bank temporary exemption.

RITS / FSS and synchronisation

Austraclear uses a proprietary direct RITS network link. PEXA and ASX Financial Settlements (for Sympli) submit RITS reservation batches. FSS settles immediately or rejects, 24/7, without reservations. HVCS governs payments; RITS settles reserves.

Consultation / legislative context

The September RITS consultation covers synchronisation, token exchange, reserve access and tokenised reserves. Reforms cover clearing competition (2023), FMI (2024), payments modernisation (2025) and the digital-assets framework (2026, effective April 2027).

Domestic public
Domestic industry / user-owned
Domestic commercial
External industry / user-owned
External corporate

Acronyms

4CB — CNAPS2

4CB Four central banks: Germany, France, Italy, and Spain • **ABS** Association of Banks in Singapore • **ACCC** Australian Competition and Consumer Commission • **ACH** Automated clearing house • **ACSS** Automated Clearing Settlement System • **ADI** Authorised deposit-taking institution • **AGDIS** Australian Government Digital ID System • **AmEx** American Express • **AML / CFT** Anti-money laundering / combating the financing of terrorism • **AMM** Automated market maker • **ANBIMA** Brazilian Financial and Capital Markets Association • **APP** Australian Payments Plus • **API** Application programming interface • **ASX** Australian Securities Exchange • **AusPayNet** Australian Payments Network • **AUSTRAC** Australian Transaction Reports and Analysis Centre • **AWS** Amazon Web Services • **B3** Brasil, Bolsa, Balcão; Brazilian exchange group • **Bacs** Bankers' Automated Clearing Services • **BaNCS** Tata Consultancy Services' banking and securities software • **BCB** Banco Central do Brasil • **BC Card** Korean bank-card processing company • **BCS** Banking Computer Services • **BECS** Bulk Electronic Clearing System • **BEPS** Bulk Electronic Payment System • **BIS** Bank for International Settlements • **BME** Bolsas y Mercados Españoles • **BNY** Bank of New York Mellon • **BoC** Bank of Canada • **BOCHK** Bank of China (Hong Kong) • **BoE** Bank of England • **BOJ / BOJ-NET** Bank of Japan / its Financial Network System • **BOK / BOK-Wire+** Bank of Korea / its interbank funds-transfer system • **BPAY** Australian electronic bill-payment service • **BT** British Telecommunications • **BX** Berne exchange; former BX Digital name • **C7** Euronext's clearing-system name • **CBDC / wCBDC** Central bank digital currency / wholesale CBDC • **Cboe** Chicago Board Options Exchange; original name • **CBUAE** Central Bank of the United Arab Emirates • **CCASS** Central Clearing and Settlement System • **CCDC** China Central Depository & Clearing • **CCP** Central counterparty • **CDCC** Canadian Derivatives Clearing Corporation • **CDP** Central Depository (Singapore) • **CDS / CDSX** Canadian Depository for Securities / its clearing and settlement system • **Cetip** Central de Custódia e de Liquidação Financeira de Títulos • **CFFEX** China Financial Futures Exchange • **CHAPS** Clearing House Automated Payment System • **CHATS** Clearing House Automated Transfer System • **CHESS** Clearing House Electronic Subregister System • **CHIPS** Clearing House Interbank Payments System • **CIPS** Cross-border Interbank Payment System • **CLS** Continuous Linked Settlement • **CME** Chicago Mercantile Exchange • **CMU** Central Moneymarkets Unit • **CNAPS2** China National Advanced Payment System, second generation

CNCC — IBM

CNCC China National Clearing Center • **CNFN** China National Financial Network • **COIN** Community of Interest Network • **CORE** Compensation REtail; STET's retail clearing platform • **CPMI** Committee on Payments and Market Infrastructures • **CREST** Euroclear UK & International's securities settlement system • **CRNet** Swift's CREST connectivity service • **CSD** Central securities depository • **CSN** Canadian Payments Association Services Network • **DAP** Digital Asset Platform; separate Goldman Sachs and SIX platforms • **DBS** Development Bank of Singapore; now DBS Bank • **DCJPY** Digital Currency JPY; Japanese bank deposit-token platform • **DeFi** Decentralised finance • **DIGIT** Digital Gilt Instrument • **DLT** Distributed ledger technology • **DSD** Digital Securities Depository • **D-SIB** Domestic systemically important bank • **DTC** Depository Trust Company • **DTCC** Depository Trust & Clearing Corporation • **DvP / PvP** Delivery versus payment / payment versus payment • **EBA** Euro Banking Association; origin of EBA CLEARING • **EBICS** Electronic Banking Internet Communication Standard • **ECB** European Central Bank • **eftpos** Electronic funds transfer at point of sale • **Embratel** Empresa Brasileira de Telecomunicações • **EN** Electronic Payments Network • **ERC** Ethereum Request for Comments; token-standard series • **ESMIG** Eurosystem Single Market Infrastructure Gateway • **EURO1** EBA CLEARING's large-value euro payment system • **EXEGO** Australexa's securities settlement-system name • **FAPI** Financial-grade API • **FAST** Fast and Secure Transfers • **FDIC** Federal Deposit Insurance Corporation • **Fed** Federal Reserve • **FedACH** Federal Reserve automated clearing house service • **FGIS** Financial Gateway Information Services • **FICC** Fixed Income Clearing Corporation • **FMI** Financial market infrastructure • **£FnPS** Sterling Faslty Payment System • **FPS** Faster Payments Service (UK); Faster Payment System (Hong Kong) • **FSS** Fast Settlement Service • **FX** Foreign exchange • **FXYCS** Foreign Exchange Yen Clearing System • **GENIUS** Guiding and Establishing National Innovation for U.S. Stablecoin Act • **GFT** Gesellschaft für Technologietransfer; technology supplier • **GIRO** General Interbank Recurring Order • **HELOC** Home equity line of credit • **HKCC** Hong Kong Futures Exchange Clearing Corporation • **HKEX** Hong Kong Exchanges and Clearing • **HKICL** Hong Kong Interbank Clearing Limited • **HKMA** Hong Kong Monetary Authority • **HKMC** Hong Kong Mortgage Corporation • **HKSCC** Hong Kong Securities Clearing Company • **HM** His Majesty's; in HM Treasury • **HSBC** Hongkong and Shanghai Banking Corporation • **HVCS** High Value Clearing System • **HVPS** High Value Payment System • **IBM** International Business Machines

IBPS — RT1

IBPS Internet Banking Payment System • **ICE** Intercontinental Exchange • **ICS** Image Clearing System • **IMN** Inter-Member Network • **IOSCO** International Organization of Securities Commissions • **ISO** International Organization for Standardization • **IT** Information technology • **JASDEC** Japan Securities Depository Center • **JCB** Japan Credit Bureau; card-network name • **JDCC** JASDEC DVP Clearing Corporation • **JGB** Japanese government bond • **JPX** Japan Exchange Group • **JPYC** Japanese yen stablecoin issued by JPYC Inc. • **JSCC** Japan Securities Clearing Corporation • **JSON** JavaScript Object Notation • **JV** Joint venture • **KFTC** Korea Financial Telecommunications & Clearings Institute • **Koscom** Korea Securities Computing Corporation; original name • **KRX** Korea Exchange • **KSD** Korea Securities Depository • **KT** Korea Telecom; now KT Corporation • **KYC** Know your customer • **L1** Layer 1 blockchain • **LCH** London Clearing House; original name • **LG CNS** LG group technology-services company • **LINK** UK cash-machine network • **LME** London Metal Exchange • **LSEG** London Stock Exchange Group • **LSEG B3** LSEG's UK digital securities-depository company • **MAS** Monetary Authority of Singapore • **MBS** Mortgage-backed securities • **MEPS+** MAS Electronic Payment System, enhanced version • **MiCA** Markets in Crypto-Assets Regulation • **MTR** Mass Transit Railway • **Nasdaq** National Association of Securities Dealers Automated Quotations; original name • **NBB-SSS** National Bank of Belgium Securities Settlement System • **NET** Messaging network (role label) • **NETS** Network for Electronic Transfers • **NFT** Non-fungible token • **NPP / NPPA** New Payments Platform / NPP Australia • **NSCC** National Securities Clearing Corporation • **NSS** National Settlement Service • **NTT Data** Nippon Telegraph and Telephone group's data-services company • **NUCC** NetUnion Clearing Corporation • **OCBC** Oversea-Chinese Banking Corporation • **OCC** Options Clearing Corporation • **OCC (US bank regulator)** Office of the Comptroller of the Currency • **ODP** Orion Derivatives Platform • **ODX** Osaka Digital Exchange • **OP / TECH** Operator / core technology supplier (role labels) • **OTC** Over the counter • **PBOC / PBoC** People's Bank of China • **PDS** Payment Delivery System • **PEXA** Property Exchange Australia • **QBFT** Byzantine-fault-tolerant consensus protocol used by Besu • **R3** Company behind the Corda distributed-ledger platform • **RBA** Reserve Bank of Australia • **RITS** Reserve Bank Information and Transfer System • **RLN** Regulated Liability Network • **RNI** Rete Nazionale Interbancaria; Italian interbank network • **RSFN** Rede do Sistema Financeiro Nacional; Brazilian financial-system network • **RT1** EBA CLEARING's instant euro payment system

RT2 — Zengin

RT2 Bank of England's renewed RTGS service • **RTGS** Real-time gross settlement • **RTM** Rede de Telecomunicações para o Mercado • **RTP** Real-Time Payments; The Clearing House's instant-payment network • **RWA** Real-world asset • **SA / S.p.A.** Société anonyme / società per azioni; company legal forms • **SCHA** Singapore Clearing House Association • **SCION** Scalability, Control, and Isolation On Next-generation Networks • **SDNet/2** HKEX's second-generation Securities and Derivatives Network • **SDX** SIX Digital Exchange • **SECOM** Settlement Communication System • **Selic** Sistema Especial de Liquidação e de Custódia • **SEOCH** Stock Exchange of Hong Kong Options Clearing House • **SETTLE** Settlement institution or system (role label) • **SGX / SGX-DC** Singapore Exchange / SGX Derivatives Clearing • **SHCH** Shanghai Clearing House • **SIA / SIAnet** Società Interbancaria per l'Automazione / its network, now Nexi • **SIC** Swiss Interbank Clearing • **SILOC** Deferred settlement system for interbank credit-order transfers • **SIS** SEGA Intersettle; SIX's securities-depository business • **SITRAF** Sistema de Transferência de Fundos • **SIX** Swiss Infrastructure and Exchange • **SMART** Securely Managed and Reliable Technology; DTCC network • **SNB** Swiss National Bank • **SPaN** Singapore Payments Network • **SPI** Sistema de Pagamentos Instantâneos • **SSFN** Secure Swiss Finance Network • **StanChart** Standard Chartered • **START** Osaka Digital Exchange's security-token market • **STEP2** Straight Through Euro Payments; EBA CLEARING's retail system • **STET** Systèmes Technologiques d'Echange et de Traitement • **STR / STR-Web** Sistema de Transferência de Reservas / its web access • **SWIAT** Secure Worldwide Interbank Asset Transfer • **Swift / SwiftNet** Society for Worldwide Interbank Financial Telecommunication / its network • **T2** Eurosystem's wholesale real-time settlement service • **T2S** TARGET2-Securities • **TARGET** Trans-European Automated Real-time Gross settlement Express Transfer • **TCH** The Clearing House • **TCS** Tata Consultancy Services • **TFX** Tokyo Financial Exchange • **TIPS** TARGET Instant Payment Settlement • **TMX** Canadian exchange group; Toronto and Montréal exchanges • **TNS** Transaction Network Services • **TSE** Tokyo Stock Exchange • **TWINT** Swiss mobile-payment service name • **UBS** Swiss bank; historically Union Bank of Switzerland • **uMINT** UBS USD Money Market Investment Fund Token • **UOB** United Overseas Bank • **USBE** United States Bulk Exchange • **WA** Western Alliance • **Zengin / Zengin-Net** Japan's Domestic Funds Transfer System / Japanese Banks' Payment Clearing Network

Currency codes AUD Australian dollar; CAD Canadian dollar; CHF Swiss franc; CNY / RMB Chinese yuan / renminbi; EUR euro; GBP pound sterling; HKD Hong Kong dollar; JPY Japanese yen; KRW South Korean won; SGD Singapore dollar; USD US dollar; BRL Brazilian real; e-CNY digital renminbi. **Country codes** BE Belgium; DE Germany; IT Italy; LU Luxembourg; NL Netherlands; UK United Kingdom; US United States.